

Managing Complicatedness in Knowledge Work: How Deterministic and Probabilistic Information Systems Shape Performance^{*}

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Abstract

Organizations deploy information systems to help knowledge workers manage portfolios of interconnected products and customers. Drawing on complexity science, we distinguish portfolio complicatedness, the number of interconnected components a worker carries, from complexity, which emerges when those interactions become ad hoc and nonlinear. We theorize that deterministic CRM features impose predefined structure keeping complicated portfolios tractable, whereas probabilistic GenAI features generate context-dependent outputs that fuel complexity dynamics. Using agent- and deal-level data from U.S. independent insurance distribution and a shift-share identification strategy, we find the opposite. CRM use negatively moderates the performance effects of product and customer complicatedness, while GenAI use positively moderates both. Predetermined structure is valuable when complicatedness creates coordination and retrieval demands, but constrains performance when those demands remain tractable. The findings extend complexity science by showing that information systems improve performance both by taming complexity and by increasing worker capacity under demanding interaction conditions.

Keywords: *complexity science, complicatedness, knowledge work, deterministic enterprise systems, probabilistic AI systems, generative AI, CRM*

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1 Introduction

Knowledge workers increasingly manage portfolios containing many products, customers, and interdependent activities. Greater portfolio breadth can create value because it enlarges the set of products that can be matched to customer needs, the range of customer situations from which a worker can learn, and the opportunities for recombination across cases (Johnson & Sohi, 2014; Weitz et al., 1986). Yet the same breadth increases the information, dependencies, and contingencies that must be represented and acted upon (Schmitz & Ganesan, 2014). Information systems are expected to help workers manage these demands (Ahearne et al., 2007; Jayachandran et al., 2005), but their value depends on how they alter the relationship between portfolio variety and the worker's capacity to respond to it (Ko & Dennis, 2011).

This problem has often been discussed in IS research under the broad label of complexity. Much of that literature, however, uses complexity to describe what more recent complexity-science-informed work distinguishes as complicatedness: the number, variety, heterogeneity, or interdependence of components in a system. Mocker et al. (2014), for example, define business complexity in terms of variety and links and show that high product complexity can be value creating when firms prevent the associated processes from becoming unnecessarily burdensome. In the terminology adopted here, much of this product complexity is better understood as product complicatedness. Tanriverdi and Du (2020) and Tanriverdi et al. (2025) sharpen the distinction by separating proliferation of interconnected components from complexity that emerges through the form and dynamics of interactions among those components. The distinction matters because increasing complicatedness creates conditions under which complexity may emerge, but does not establish that it has emerged or that additional variety is harmful.

Recent IS research has made an important advance by showing how technology can tame complexity (Benbya et al., 2020). In multihospital systems, for example, enterprise-wide data analytics platforms replace ad hoc cross-hospital information exchange with standardized, controlled flows, thereby attenuating the cybersecurity consequences of service, IT, and governance complicatedness (Tanriverdi et al., 2025). This mechanism is theoretically compelling at the organizational-system

level: hospitals are constituent units, cross-hospital exchanges are interactions among those units, and an enterprise platform changes the architecture through which those interactions occur. Yet transferring this logic to individual knowledge work embeds a consequential assumption - that structuring an individual's information and action becomes increasingly valuable as the individual's task portfolio becomes more complicated.

We problematize that cross-level extrapolation. The components of an individual knowledge worker's task portfolio do not interact autonomously in the same way organizational units do. Their dependencies are interpreted and acted upon through a focal worker. An IS therefore need not only restructure interactions among system components; it can change how the worker represents, searches, interprets, and responds to the variety those components create. This difference is especially important when complicatedness is potentially valuable. [Mocker et al. \(2014\)](#) show that firms may deliberately retain valuable product variety while simplifying the processes used to handle it, and they also describe tools and roles that enable employees to perform difficult processes without eliminating the underlying variety. [Nan and Tanriverdi \(2017\)](#), drawing on requisite variety, similarly show that information alone becomes insufficient when the variety of stimuli exceeds the response variety available to an agent and identify generativity as a means of expanding response variety. Together, these arguments suggest that the relevant question at the individual level is not simply how technology suppresses interaction variety, but how it changes the worker's capacity to exploit valuable variety without being overwhelmed by its demands.

We examine two contrasting classes of IS feature use within the same work platform. A deterministic feature returns the same output given the same inputs and configuration. A probabilistic feature produces model-generated output that may vary across identical invocations and that the worker accepts, edits or reverts before use ([Benbya et al., 2024](#)). Deterministic CRM features represent customers, opportunities, products, and activities through predefined records, categories, states, and workflows. Such structure supports retrieval and coordination, but it also requires heterogeneous situations to pass through representations specified in advance. As portfolio complicatedness rises, the number of contingencies that must fit those representations grows, increasing

the potential for representational compression, administrative burden, and displacement of situational adaptation ([Strong & Volkoff, 2010](#); [Tarafdar et al., 2015](#); [Vieira da Cunha et al., 2015](#)). We therefore expect the marginal performance return to complicatedness to decline as deterministic CRM Use increases.

Probabilistic GenAI features operate differently. Generated outputs are context dependent and require worker evaluation, so they can introduce verification and revision costs ([Alavi et al., 2024](#); [Becker et al., 2025](#)). But probabilistic generation also expands the set of candidate interpretations and responses available without requiring each to have been specified *ex ante*. This property should matter increasingly as task variety rises. When the worker's existing repertoire is adequate, generative assistance can add cost without much additional capacity. As product and customer variety expand, however, the worker encounters more configurations for which existing rules, scripts, and representations are incomplete. GenAI can then increase response variety through synthesis, recombination, and generation of context-contingent candidates ([Brynjolfsson et al., 2025](#); [Nan & Tanriverdi, 2017](#)). We therefore expect GenAI Use to increase the marginal performance return to complicatedness.

We test these arguments in U.S. independent insurance distribution. The focal work system is an individual agent's deal portfolio, and the unit of analysis is the agent-week. Product categories and customer segments constitute distinct components because agents must match heterogeneous products to heterogeneous customer needs while carrying prior commitments and knowledge across concurrent relationships. Using approximately 312,000 deal records and digital traces of deterministic CRM and probabilistic GenAI feature use, we estimate how product and customer complicatedness relate to conversion and how each class of feature use moderates those relationships. Agents are assigned their portfolios of work, and their use of each class of feature responds to the portfolio they carry. The empirical design instruments complicatedness, CRM Use, and GenAI Use to recover the causal moderation, estimating it through a control-function specification with correlated random coefficients.

The findings support the two hypotheses. CRM Use negatively moderates the relationship

between complicatedness and conversion for both product and customer complicatedness, whereas GenAI Use positively moderates both relationships, less precisely so at high customer complicatedness. The baseline shapes are theoretically important: customer complicatedness initially raises conversion before reaching a turning point, while product complicatedness remains performance enhancing throughout the observed range, although at a diminishing rate. Thus, complicatedness is not intrinsically harmful in this setting. The nonlinear CRM pattern adds a further boundary condition: although CRM reduces the returns to customer complicatedness across much of the range, it becomes relatively protective toward the upper end where additional customer complicatedness itself begins to reduce conversion.

The study contributes to IS complexity research in three ways. First, it clarifies a construct distinction obscured by broad uses of the term complexity. Prior studies that operationalize variety, number, heterogeneity, or links among components primarily inform the antecedent condition we call complicatedness; studies of contingent, adaptive, ad hoc, nonlinear, or emergent interactions more directly address complexity. Second, it identifies level of analysis as a boundary condition on technological complexity management. Enterprise systems can tame complexity by restructuring interactions among organizational units; individual-use systems can instead alter how a focal actor represents and responds to task variety. Third, it distinguishes two technological responses to rising complicatedness: predetermined structuring, which constrains the interaction and representation space, and response-variety augmentation, which expands the actor's repertoire of context-contingent responses. The findings show that their relative value depends on whether additional variety remains productively exploitable or has begun to exceed the worker's capacity to manage it.

2 Theoretical Foundation

2.1 Complicatedness and Complexity in IS Research

Complexity science distinguishes a system's constituent variety from the dynamics that arise through interaction. A complicated system contains numerous interconnected components whose relationships remain sufficiently structured to be analyzed and coordinated. Complexity is different: it emerges when local interactions, feedback, adaptation, and interdependence produce system behavior that cannot be understood by treating components independently ([Benbya et al., 2020](#); [Cilliers, 1998](#); [Holland, 1995](#)). The distinction is analytical rather than merely semantic. Adding components necessarily increases the number of elements and potential dependencies that must be managed, but it does not necessarily make their interactions ad hoc, nonlinear, or emergent ([Miller & Page, 2007](#)).

IS research has not always maintained this distinction. Many studies use complexity as an umbrella term for component number, product variety, organizational differentiation, interconnectedness, or processing burden ([van den Hooff et al., 2026](#)). Those studies remain theoretically important, but their arguments should be interpreted according to what they conceptualize and measure rather than according to the label alone. [Mocker et al. \(2014\)](#), for example, define business complexity through variety and links across products, processes, and organizational arrangements. Their finding that product complexity can be rewarding therefore provides evidence that greater component variety and interconnection - complicatedness in our terminology - can create value. It does not require the stronger conclusion that ad hoc or nonlinear emergent dynamics are themselves beneficial.

Recent IS work makes the distinction more explicit. [Tanriverdi and Du \(2020\)](#) conceptualize multibusiness firms as systems of business units and distinguish changes that alter the number or composition of businesses from changes that alter interrelatedness among them. Their longitudinal evidence shows that strategy changes involving related businesses increase the hazard of IT control material weaknesses, whereas changes involving unrelated businesses do not have comparable

effects. [Tanriverdi et al. \(2025\)](#) go further by distinguishing the richness of interaction from its form. In multihospital systems, service, health IT, and governance complicatedness increase cybersecurity risk, but an enterprise-wide data analytics platform attenuates these effects by converting ad hoc cross-unit data exchanges into more structured and controlled flows.

[Nan and Tanriverdi \(2017\)](#) illuminate a different part of the complexity problem. Their agent-based model is explicitly multilevel: firm-level IT-enabled strategic actions and interactions aggregate into collective-level hyperturbulence, which then feeds back to firm-level opportunities for competitive advantage. The simulation therefore models complexity as a process of cross-level emergence rather than as component count. Their analysis also distinguishes reducible uncertainty, for which better information can help an agent select among existing responses, from conditions in which the agent lacks sufficient response variety. Under the latter condition, generativity can increase the repertoire available for responding to environmental variety. This distinction becomes central when the focal system is an individual knowledge worker rather than an organization composed of interacting subunits.

[Table 1](#) makes these differences explicit. We preserve each study's original terminology while separately interpreting what its constructs and measures capture under the complicatedness-complexity distinction adopted here. The purpose is not to relabel prior work retrospectively, but to prevent construct labels from obscuring differences in system boundary, interaction level, and outcome.

2.2 Complicatedness as Opportunity and Demand

Once complicatedness is separated from complexity, there is no theoretical reason to treat additional components as intrinsically harmful. Additional variety can create opportunities for matching, recombination, learning, and differentiation while simultaneously increasing the demands required to exploit those opportunities. [Mocker et al. \(2014\)](#) make this point directly in their analysis of the complexity sweet spot: firms can benefit from extensive product variety and links when the processes used to deliver that variety remain manageable. In our terminology, complicatedness can

Table 1. Selected empirical and computational IS studies of complicatedness and complexity

Study	Authors' term	What is conceptualized / measured	System & level	Role of IS / interaction mechanism	Outcome
Xia and Lee (2005)	ISD project complexity	Structural and dynamic complexity in organizational and IT domains; the structural terms correspond to complicatedness	IS development project; project	The project is the system; no technology moderates it	Project duration; undesirable
Mocker et al. (2014)	Product and process complexity	Variety and links; much of the construct corresponds to complicatedness under the present distinction	Firm; product and process	Digitization can preserve valuable product variety while simplifying processes or making difficult processes easier to execute	Firm performance; desired
Liang et al. (2015)	System complexity	Perceived difficulty of understanding and using an enterprise system; anchored on Simon's non-simple interaction	Enterprise system in use; individual	System complexity moderates the effects of job autonomy and task variety on system exploration	System exploration and extended use; desired
Nan and Tanriverdi (2017)	Hyperturbulence / complexity dynamics	Nonlinear collective dynamics generated by recurrent firm actions and interactions; response variety and generativity	Firms nested in competitive collective; multilevel	Component and architectural IT alter productive means, interconnections, information, and generativity	Firm competitive advantage and collective hyperturbulence
Tanriverdi and Du (2020)	Complicatedness and complexity structure	Business-unit composition versus interrelatedness among business units	Multibusiness firm; organizational	IT controls operate at cross-business integration points affected by strategy changes	IT control material weaknesses; undesirable
Flyvbjerg et al. (2022)	Component interdependence in IT systems	Power-law distribution of cost overruns; complexity inferred from the distribution rather than from a count	IT project; project	Interdependence among components under self-organized criticality generates the distribution; shown by simulation	Cost overrun; undesirable
Liang et al. (2025)	Complexity from mergers and acquisitions	Annual count of acquisitions and parent-target dissimilarity; both treated as sources of nonlinear complexity	US public firms, acquiring and non-acquiring; organizational	Structure, process and technology joined by acquisition raise exposure; IT integration argued, not measured	Data breaches; undesirable
Tanriverdi et al. (2025)	Complicatedness and complexity	Service, health IT, and governance complicatedness; ad hoc versus structured information sharing	Multihospital system; organizational / system	EWDAP restructures cross-hospital information exchange into standardized, controlled flows	Cybersecurity breaches; undesirable
van den Hooff et al. (2026)	Structural, dynamic, and subjective complexity	Size, multiplicity and interdependencies of components, separated from rate of change and from perception; the structural form is complicatedness	IT architecture; organizational	Modularisation choices shift complexity within and between forms	Manageability; mixed

Note. The table preserves the terminology used by each study while distinguishing the underlying construct using the definitions adopted in this paper. "Complicatedness" refers to proliferation, variety, or interconnection of components; "complexity" refers more directly to contingent, adaptive, nonlinear, or emergent interaction dynamics.

therefore be productive even though it increases the conditions under which difficult interaction dynamics may arise.

For an individual knowledge worker, complicatedness increases the variety of states that must be recognized and the conditional responses that may be required (Weitz et al., 1986). Product complicatedness expands the set of product rules, eligibility conditions, exclusions, documentation requirements, and possible product-to-customer mappings (Johnson & Sohi, 2014). Customer complicatedness expands the set of needs, histories, stakeholder structures, communication expectations, and relational contingencies the worker must interpret (Schmitz & Ganesan, 2014). These demands can fragment attention and dilute specialized learning, but they can also enrich the worker's repertoire through comparison, transfer, and recombination across cases (Aral et al., 2012; Eppler & Mengis, 2004; Spiro & Weitz, 1990; Weitz et al., 1986).

The performance consequence of complicatedness therefore depends on the relationship between task variety and response variety. Complex adaptive systems theory treats agents as systems that transform information from their environments into action through behavioral rules that can be retained, revised, or recombined as feedback arrives (Arthur, 2015; Holland, 1995). Ashby's principle of requisite variety implies that effective regulation requires a response repertoire commensurate with the variety of states that must be handled (Ashby, 1956). Nan and Tanriverdi (2017) apply this reasoning in IS strategy: information can reduce uncertainty when an agent already possesses an adequate response, but additional information is insufficient when the environment presents stimuli for which the agent lacks appropriate responses. Generativity then matters because it expands possible responses.

This reasoning also clarifies the transition from complicatedness toward complexity. A worker can manage many components without complex dynamics when established representations and responses remain adequate. As task variety increasingly outruns the worker's available repertoire, however, interactions must be reconstructed case by case (Holland, 1995). Responses become more contingent on the particular combination of product rules, customer conditions, prior actions, and feedback. Complicatedness thus raises the opportunity for complexity to emerge without implying

that complexity is directly observed whenever component breadth rises.

2.3 Level of Analysis and the Role of Information Systems

The appropriate technological response to complicatedness depends on where the focal interactions occur. At the organizational-system level, constituent units can exchange information and resources with one another through an architecture that is itself a target of intervention. [Tanriverdi et al. \(2025\)](#) illustrate this mechanism clearly. Member hospitals use heterogeneous systems and governance arrangements, and ad hoc cross-hospital data exchange creates security vulnerabilities. EWDAP introduces common formats, repositories, protocols, access rules, and interfaces. The technology therefore changes the interaction architecture among organizational components, making cross-unit exchanges more standardized and controllable.

Circumscribing the boundary of a complex system is one of the challenges complexity science poses for IS research ([Benbya et al., 2020](#)). At the individual level, the causal locus is different. Products and customers in a task portfolio create dependencies and information demands, but a focal worker interprets and acts upon those demands. The relevant IS intervention is therefore not limited to changing connections among autonomous subunits. It can change what the worker sees, how situations are represented, which prior information can be retrieved, which actions are prescribed, and which candidate responses can be generated. A mechanism that is beneficial because it constrains cross-unit interactions at the organizational level does not automatically imply that progressively constraining an individual's information and action is beneficial as task variety grows.

This cross-level distinction exposes two ways IS can help an individual manage complicatedness. One is interaction structuring: technology reduces the variety the worker must actively reconstruct by fitting information and action into predefined representations, states, and workflows. The other is response-variety augmentation: technology increases the range of interpretations and responses the worker can produce for heterogeneous situations. These mechanisms are not mutually exclusive properties of entire platforms. They describe feature-level interaction logics that can coexist within

the same system. Deterministic CRM features primarily instantiate the first logic; probabilistic GenAI features can instantiate the second.

The distinction also changes the objective of complexity management. In cybersecurity research, the focal outcome is an undesirable system-level event, and restraining ad hoc interaction can directly reduce pathways to failure. In knowledge work, the focal actor intentionally seeks to generate a desired outcome. Conversion requires the worker to exploit heterogeneity by constructing a product-customer response that fits a particular situation. The objective is therefore not to minimize variety as such, but to preserve valuable variety while keeping the demands it creates within the worker's effective response capacity.

3 Hypotheses Development

3.1 Deterministic CRM Features: Predetermined Structure and Increasing Task Variety

Deterministic CRM features manage complicatedness by imposing a predefined representational and procedural structure. Customer histories, opportunity states, product categories, activities, documents, and next actions are captured in fields, stages, records, and workflows. This structure has clear benefits. Durable records reduce reconstruction from memory, sequencing supports coordination across concurrent opportunities, and retained histories preserve continuity across relationships (Ahearne et al., 2007; Alavi & Leidner, 2001; Jayachandran et al., 2005; Ko & Dennis, 2011; Reinartz et al., 2004). At a fixed level of task variety, these functions can make work more visible, retrievable, and governable.

The individual-level question, however, is how the marginal value of that predetermined structure changes as the portfolio becomes more complicated. Every additional product category and customer segment introduces situations that may differ in the information that matters, the appropriate sequence of action, and the response required. Predefined representations necessarily compress those particulars into categories established before the focal situation occurs. As variety grows, a larger set of contingencies must be represented through the same field structures and workflow logic.

Representational fit therefore becomes harder to maintain. Enterprise-system research shows that standardized representations can misfit local work, while CRM research emphasizes that relational information processes create value when they capture what is particular to the customer rather than merely impose uniform recording (Boulding et al., 2005; Jayachandran et al., 2005; Strong & Volkoff, 2010).

The administrative cost of predetermined structure also rises with the component set. Each additional active component generates information that must be entered, classified, updated, and maintained. These obligations are incurred whether the focal situation actually requires additional ordering. As task variety increases, CRM Use can therefore substitute recording and compliance effort for adaptive work (Vieira da Cunha et al., 2015), a possibility consistent with technostress research on technology-enabled administrative burden (Tarafdar et al., 2015). The problem is not that structure has no value, but that its constraints scale with component breadth while its coordination benefits depend on whether the added variety actually requires the same predefined structure.

This argument differs from organizational-level complexity taming. EWDAP can improve an MHS by changing the protocols through which hospitals exchange data; its standardization acts directly on interactions among organizational components (Tanriverdi et al., 2025). CRM Use by an individual instead acts through the representation and handling of heterogeneous task states. When additional task variety remains potentially valuable, increasingly fitting that variety to predefined representations can restrict the worker's ability to exploit differences among products and customers (Li et al., 2023). We therefore expect the marginal performance return to complicatedness to decrease with deterministic CRM Use.

Hypothesis 1. *The use of deterministic CRM features negatively moderates the relationship between deal-portfolio complicatedness and knowledge-worker performance, such that the marginal performance return to complicatedness decreases as CRM Use increases.*

3.2 Probabilistic GenAI Features: Response-Variety Augmentation

Probabilistic GenAI features confront increasing task variety through a different interaction logic. Rather than requiring every response to be encoded in a predefined category or workflow, GenAI can generate context-dependent summaries, drafts, explanations, alternatives, and candidate responses from the information supplied in a work episode (Benbya et al., 2024). Generated output is not self-validating: the worker must judge whether it fits relevant product rules, customer conditions, relationship histories, and task states (Alavi et al., 2024). GenAI can therefore introduce prompting, verification, revision, and grounding costs (Becker et al., 2025; Dell’Acqua et al., 2026; Hohenstein et al., 2023; Susarla et al., 2023).

Those costs imply that GenAI need not be valuable when complicatedness is low. When a worker confronts a small and familiar set of products and customer situations, existing knowledge, routines, and deterministic records may already supply adequate responses (Brynjolfsson et al., 2025). Generating and checking an additional candidate then adds an interaction without necessarily expanding effective capacity (Fügener et al., 2021; Luo et al., 2021). The value of probabilistic generation should change, however, as the variety of situations exceeds what the worker can readily maintain through existing scripts and representations.

Requisite-variety reasoning explains why. As environmental variety rises, effective action requires a correspondingly varied response repertoire (Ashby, 1956). Nan and Tanriverdi (2017) distinguish uncertainty that can be reduced by supplying information from conditions in which the agent lacks sufficient response variety; they identify generativity as a mechanism through which IT can expand the set of possible responses. At the individual level, GenAI can perform a related function by lowering the cost of constructing candidate interpretations and responses for configurations that were not exhaustively anticipated (Doshi & Hauser, 2024; Holland, 1995). It can recombine product information, customer context, and prior interaction into alternatives that the worker evaluates rather than requiring the worker to retrieve a fully formed response from an existing repertoire (Althuizen & Wierenga, 2014).

The marginal value of this capacity should increase with portfolio complicatedness. More

product categories create more combinations of rules, exclusions, and use cases that may need synthesis (Johnson & Sohi, 2014; Mocker et al., 2014). More customer segments create more heterogeneous histories, preferences, stakeholder structures, and communication requirements that may require adaptation (Schmitz & Ganesan, 2014). Probabilistic generation can increase the worker's response variety precisely where predefined responses become less complete. It does not eliminate the need for judgment and may itself create contingent interaction paths, but those costs can be outweighed by the additional adaptive capacity supplied when task variety is high (Cilliers, 1998). We therefore expect GenAI Use to increase the marginal performance return to complicatedness.

Hypothesis 2. *The use of probabilistic GenAI features positively moderates the relationship between deal-portfolio complicatedness and knowledge-worker performance, such that the marginal performance return to complicatedness increases as GenAI Use increases.*

4 Methodology

4.1 Empirical Setting

Independent insurance sales is a knowledge-intensive setting in which agents manage portfolios of opportunities that span multiple product categories and customer segments. An agent's task portfolio takes the form of a deal portfolio in this setting, and that deal portfolio is the focal work system in our theory. Product categories and customer segments are components of that system. An agent may combine several products to construct coverage for a single customer, and differences among customers require the agent to determine which prior configurations, knowledge, and interactions can be reused and which must be adapted. Increasing the number of distinct product categories or customer segments therefore increases the complicatedness of the portfolio. The setting also allows us to observe deterministic CRM-feature use and probabilistic GenAI-feature use within the same platform and by the same worker.

The components of these portfolios are substantively distinct. Agents represent broad ranges of property and casualty products, from personal auto and homeowners' policies to commercial general liability, workers compensation, and cyber insurance, each governed by its own coverage structures, pricing logic, underwriting rules, and regulatory requirements. They serve equally diverse customers, including individual consumers, small and medium-sized businesses, agricultural enterprises, financial services firms, and industries with specialized risk profiles, each requiring different engagement strategies, decision criteria, and relational approaches. Each opportunity must also be advanced through a multi-step pipeline whose procedural demands, including documentation, approvals, handoffs, and stakeholder coordination, differ depending on which product is being sold to which customer. [Figure 1](#) shows the stages of this pipeline.

[Insert [Figure 1](#) about here]

The feature-level distinction is especially clear in this channel. The platform serves independent agencies, which own and manage their customer lists while placing business with multiple insurers ([Forman & Gron, 2011](#); [Hilliard & Tennyson, 2025](#)). Customer data are therefore not pooled into a carrier-controlled enterprise suite, and regulatory and data-governance constraints place predictive insurance modeling primarily with insurers ([National Association of Insurance Commissioners, 2023](#); [U.S. Government Accountability Office, 2019](#)). Customer relationship management platforms divide into operational architectures that automate business processes and analytical architectures that model customer characteristics and behavior ([Ngai et al., 2009](#)), and the focal platform belongs to the operational class. During the study period, the focal platform contained no feature that computed a probability, score, ranking, or propensity for an agent. The probabilistic functionality observed in the data is the platform's GenAI tool. The appendix [Table A1](#) reports the platform's module groups together with the deterministic and probabilistic features each contains. Probabilistic features recur across most module groups, so the distinction separates classes of feature within the platform and classifies use at the individual logged-action level.

Our empirical analysis utilizes proprietary data¹ from a leading Insurtech software-as-a-service

¹Access to the data is governed by a signed non-disclosure agreement with the business provider. All regression analyses are estimated on the raw, unscaled data. Descriptive statistics and summary measures that could reveal

CRM provider.² The study draws on roughly 312,000 dealcard³ records in the customer acquisition pipeline during the period of study (January to July 2024). In addition to the dealcard records, we collect digital trace data⁴ capturing clickstream activity by users of the CRM. Each interaction with the CRM features and GenAI tool is logged with precise timestamps in the trace data, enabling us to identify when and how intensively an agent used the technology across different tasks over time.

4.2 Sample Construction and Variables

The sample is constructed by applying inclusion criteria at the agency, agent, and dealcard levels to ensure that observations reflect active, observable customer acquisition activity on the CRM platform. Agencies are included if they maintain an active acquisition pipeline, generate a minimum volume¹ of monthly business during the study window, and have at least a minimum number¹ of licensed agents operating on the platform. Because construction of the conversion-stage focal measures and their qualification-stage controls requires distinguishing earlier from later pipeline activity, we further require each agency's pipeline to exhibit at least a three-stage funnel structure; agencies whose pipelines lack sufficient stage differentiation are excluded.

Conversion. Sales performance is captured using conversion, the rate at which active later-stage

commercially sensitive scale or operational details are reported only after applying a confidential, constant linear scaling factor that preserves the variation used in empirical analysis while preventing recovery of proprietary magnitudes, consistent with prior work using proprietary or confidentially governed business data (Fontnouvelle et al., 2007; Nair et al., 2017; Yoganarasimhan et al., 2023). Furthermore, specific thresholds used cannot be disclosed under the NDA.

²The CRM provider serves small and medium independent insurance agencies across the United States. It manages premiums exceeding \$8 billion and more than 1,000 agencies nationwide use the CRM to manage their sales activities and customer relationships across diverse property and casualty insurance portfolios, and diverse groups of customers.

³A dealcard, equivalently a sales opportunity (Hartmann et al., 2025), is a centralized record serving as the core unit of analysis for salespeople and CRM system developers. It functions as a data repository for each deal, including customer details, the product or service offered, pricing, and process updates. It also records details about the agents and agencies involved, user interactions, associated accounts, and linked insurance policies. The dealcard collects timestamped traces as it moves through various stages, from initial contact and proposal to negotiation and closing, leaving a digital trace of a deal's history and status in the sales funnel.

⁴Trace data consist of timestamped records of user actions and interactions within information systems, capturing behaviours such as logging in or out, page views, button clicks, and other activities. From a developer's perspective, it also captures computational resources used, bugs, frictions, and modular environments affecting deployment of features and modular integrations. These data are collected unobtrusively as by-products of system usage, providing a detailed log of activities as users engage with the platform. The literature describes trace data as "residues left behind by multiple types of behavior" (Lindberg, 2020) and "logs of employee behavior," characterized by natural occurrence rather than active production for research purposes (Grover et al., 2020). In this setting, agents' path and browsing activity within the platform, together with timestamped clicks, can be attributed to specific workflow modules, allowing module-level engagement to be measured as clicks and typing time aggregated to the agent-week-module level.

deals resolve as successful policy purchases in a given agent-week, reflecting precision-oriented execution effectiveness at the point where deals close. Formally, let $\mathcal{D}_{it}^C$ denote the set of active later-stage deals for agent i in week t , and let N_{it} denote the number of those deals that convert in week t . We define sales performance as: $Y_{it}^C = \text{Conversion}_{it} = \frac{N_{it}}{|\mathcal{D}_{it}^C|}$.

Because Conversion is heavily skewed, it is log-transformed in estimation.

Portfolio complicatedness. Consistent with the theory, complicatedness captures the breadth of distinct components in an agent's deal portfolio. *Product Complicatedness* is the number of distinct product categories represented among the agent's active deals, and *Customer Complicatedness* is the number of distinct customer segments. These variables operationalize component breadth at the individual task-portfolio level. They do not measure complexity itself, nor do they identify whether interactions among portfolio components are structured, ad hoc, linear, nonlinear, or emergent. The focal measures are constructed from later-stage deals corresponding to the conversion outcome. Qualification-stage counterparts, the nonfocal complicatedness dimension, and process activity burden enter the models as controls.

Product Complicatedness. *Product Complicatedness* captures the number of distinct insurance product categories represented in an agent's active deal portfolio at a given stage of the sales funnel. Let $s \in \{Q, C\}$ index the stage of the funnel, where Q denotes qualification-stage deals and C denotes conversion-stage deals. Let p^d denote the product category associated with deal d . $\text{ProdCp}_{it}^s = |\{p_d : d \in \mathcal{D}_{it}^s\}|$, $s \in \{Q, C\}$. Thus, ProdCp_{it}^Q captures qualification-stage *Product Complicatedness*, while ProdCp_{it}^C captures conversion-stage *Product Complicatedness*.

Customer Complicatedness. *Customer Complicatedness* captures the number of distinct customer segments represented in an agent's active deal portfolio at a given stage of the sales funnel. Using the same stage-specific deal set, let c^d denote the customer segment associated with deal d . $\text{CustCp}_{it}^s = |\{c_d : d \in \mathcal{D}_{it}^s\}|$, $s \in \{Q, C\}$. Thus, CustCp_{it}^Q captures qualification-stage *Customer Complicatedness*, while CustCp_{it}^C captures conversion-stage *Customer Complicatedness*.

Technology use. CRMuse_{it} is the weekly number of active minutes an agent spends interacting with deterministic platform features, derived from clickstream data. GenAI_{it} is the weekly number

of active minutes the agent spends interacting with the platform's probabilistic GenAI features, constructed from the same clickstream source on the same continuous, weekly-minutes basis. Each logged action is classified at logging as deterministic or probabilistic by the procedure the feature executes, following the feature classification in [Table A1](#), so the two measures are separated at construction and never overlap. These variables measure feature-use intensity. They do not directly measure the theoretical mechanisms of predetermined interaction structuring or response-variety augmentation; those mechanisms motivate the hypotheses and the interpretation of the moderation patterns.

Controls. Because the agent manages early-stage and later-stage deals as a coupled system, the components carried at one stage remain a concurrent condition for performance at the other. In the main specification, the focal complicatedness dimension, product or customer, is measured at the conversion stage and estimated separately. The same dimension measured at the qualification stage and the non-focal dimension at both stages enter as controls. This isolates the effect of complicatedness at the stage where the outcome is measured from complicatedness carried over from the earlier stage of the same pipeline and from other portfolio demands. Additional controls include process activity burden, portfolio value, portfolio age, bundling, and portfolio size.

Process activity burden. The empirical models also control for the average accumulated transaction and activity burden embedded in an agent's active deal portfolio. Each dealcard records the cumulative count of transactions and activities that have occurred as the deal progresses through the CRM workflow. Let A^d denote the cumulative number of transactions and activities recorded for deal d . Thus, ProcAB_{it} captures the average cumulative activity count per active deal in the agent's portfolio.
$$\text{ProcAB}_{it} = \frac{\sum_{d \in \mathcal{D}_{it}} A_d}{|\mathcal{D}_{it}|}.$$

Bundling intensity. Cross-selling and promotional-bundling behavior is an established, distinct driver of conversion performance in the sales literature ([Balachander et al., 2010](#); [Schmitz et al., 2014](#)). Bundling intensity proxies for this behavior: it captures the average number of products associated with an agent's active deals. Let B^d denote the number of products linked to deal d .
$$\text{Bundling}_{it} = \frac{\sum_{d \in \mathcal{D}_{it}} B_d}{|\mathcal{D}_{it}|}.$$

Portfolio value. Financial incentive structure is an established, direct driver of sales effort and performance in the compensation literature (Basu et al., 1985; Chung & Narayandas, 2017; Rao et al., 2021). Because compensation in this setting is commission-linked, portfolio value proxies for this incentive: higher-value portfolios imply higher potential earnings. It captures the average monetary value of an agent's active deals. Let V^d denote the raw dollar value of deal d . $\text{PortfolioValue}_{it} = \frac{\sum_{d \in \mathcal{D}_{it}} V_d}{|\mathcal{D}_{it}|}$.

Portfolio age. Deal-cycle velocity, the idea that deals remaining open longer are subject to different urgency and closing dynamics than recently opened ones, is an established construct in sales performance research (Fong & Waisman, 2025; Guesalaga & Kapelianis, 2015). Portfolio age proxies for this dynamic: it captures the mean age of opportunities in the agent's active portfolio, measured in days. For each deal, age is calculated as the number of days elapsed since the deal's creation date. Let G^d_t denote this deal-level elapsed age for deal d as of week t . $\text{PortfolioAge}_{it} = \frac{\sum_{d \in \mathcal{D}_{it}} G_{dt}}{|\mathcal{D}_{it}|}$

Portfolio size. Workload, the volume of concurrent work an individual manages, is an established driver of individual performance, both generally in the operations literature (Tan & Netessine, 2014; Xu et al., 2022) and specifically for sales conversion in the marketing literature on deal volume (Ahearne et al., 2025; Sabnis et al., 2013). Portfolio size proxies for this constraint: it captures the number of active deals in an agent's portfolio at a given point in time. Let $\mathcal{D}_{it}$ denote agent i 's set of active deals in week t . $\text{PortfolioSize}_{it} = |\mathcal{D}_{it}|$.

Having defined the sample's dependent variable, we now summarize their distributions and pairwise relationships. Table 2 reports descriptive statistics for all variables described above. Table 3 reports their pairwise correlations.

[Insert Table 2 about here]

[Insert Table 3 about here]

4.3 Econometric Model and Identification

To estimate how portfolio complicatedness and technology use relate to conversion, and whether deterministic and probabilistic feature use moderate the complicatedness-performance relationship, we use the following specification, estimated separately for each focal complicatedness dimension:

$$Y_{it} = \beta_1 \text{Complicatedness}_{it} + \beta_2 \text{Complicatedness}_{it}^2 + \beta_3 \text{GenAI Use}_{it} + \beta_4 \text{CRM Use}_{it} + \beta'(\text{structural moderation terms}) + \gamma' \mathbf{X}_{it} + \mu_i + \delta_t + \varepsilon_{it} \quad (1)$$

Y is conversion performance. The quadratic term allows the complicatedness-performance relationship to be curvilinear rather than strictly monotonic. Structural moderation terms are the interactions among Complicatedness, Complicatedness², GenAI, and CRM. X includes the nonfocal complicatedness dimension, product and *Customer Complicatedness* at the qualification stage, process activity burden, portfolio value, portfolio age, bundling, and portfolio size. The technology-by-complicatedness interactions test the empirical implications of H1 and H2: whether the marginal performance return to increasing complicatedness differs systematically with the type and intensity of feature use. They do not directly observe the theorized mechanisms. In particular, the estimates do not establish whether CRM makes underlying interactions more structured or whether GenAI expands the worker's effective response repertoire, nor do they observe whether interactions become ad hoc, nonlinear, or emergent.

An OLS estimation of complicatedness can be biased through two channels. Unobserved time-varying skill, effort, opportunity quality, or workload may affect conversion while also influencing how broad a portfolio an agent carries and how intensively the agent uses CRM or GenAI. A more capable agent may be assigned a broader portfolio, use CRM and GenAI more effectively, and convert more deals, regardless of the technology's true effect. Technology use may also respond to the expected payoff from using a technology at a particular level of complicatedness, making the moderation terms also endogenous.

Agent and week fixed effects, together with the controls in X, address only part of this problem.

Agent fixed effects absorb what is time-invariant about an agent, such as persistent skill, and week fixed effects absorb what is common to all agents in a week, such as seasonal demand or macro conditions. Neither absorbs a shock specific to one agent in one week, nor the possibility that technology use responds to complicatedness within the week rather than preceding it, so that an apparent moderation effect partly reflects reverse causality.

A design that randomly assigned each agent's complicatedness and technology use every week would eliminate both problems: randomizing complicatedness independently of use rules out reverse causality, and randomizing both independently of unobserved skill rules out the confound above. Since neither can be assigned experimentally in the field, we instead pursue a quasi-experimental design. Complicatedness, *CRM Use*, and *GenAI Use* are treated as endogenous and instrumented separately, drawing on shocks unrelated to any individual agent's performance, as a substitute for the randomization the ideal experiment would have provided. In each structural model these three are the source endogenous variables, and the nonlinear and interaction terms are constructed from them within the control-function framework.

CRM Use and *GenAI Use* are each instrumented using the exogenous-shifts shift-share framework (Adão et al., 2019; Bartik, 1991; Borusyak et al., 2022, 2025), which attributes identification to quasi-random variation in a common set of shocks while allowing agents' exposure shares to be endogenous. This design requires a level of variation below the agent: the shocks and shares are defined at the level of the platform's 65 CRM workflow modules, the discrete modular components of the CRM platform through which agents complete portfolio-related work, separately identifiable in the digital trace data by the module associated with each user action. Table A1 presents these modules condensed into ten functional groups for exposition, while all shares and shocks are constructed at the module level. Advancing a deal through the sales funnel requires the joint completion of tasks across multiple interdependent modules, consistent with an O-ring production structure (Demirer et al., 2026; Gans & Goldfarb, 2026; Kremer, 1993), in which task components are complements rather than separable inputs. Agents' exposure shares are therefore constructed to reflect their module-level workflow pattern, rather than a single undifferentiated measure of

platform use, and the shock each instrument draws on is specific at the module level, described separately for CRM and GenAI below.

For each of the endogenous variables CRM_{it} and GenAI_{it} , the instrument takes the form: $z_{it} = \sum_k w_{ik,t-1} \cdot g_{kt}$; where k indexes CRM workflow modules, $w_{ik,t-1}$ is agent i 's lagged exposure share to module k , and g_{kt} is a module-week shift. Shares are normalized so that $\sum_k w_{ik,t-1} = 1$ within each agent-week and are lagged one period to ensure exposure is predetermined relative to the contemporaneous shift. Identification under this design also rests on two conditions on the shocks themselves: dispersion⁵ and independence⁶. Both conditions are tested directly in the appendix in [Section A3.4](#).

4.3.1 Instrument for CRM Use

The shock is the module's weekly dependency load, $g_{k,t} = \sum_{v \in V_{k,t}} c_v$, the total number of coded dependencies a CRM workflow module resolves across every render of its interface on the platform in week t , where $V_{k,t}$ is the set of those renders and c_v the dependency count of render v .⁷ A module is a subsystem of the platform whose internal linkages are exercised each time its interface is assembled. Near-decomposable systems are characterised by the frequency of intra-component linkage relative to intercomponent linkage ([Simon, 1962](#)), so this quantity is the realised volume of that subsystem's internal activity over the week. It is constituted at two levels of the platform, the

⁵If the shocks are concentrated in a small number of module-weeks rather than spread across many, the instrument effectively collapses to a handful of large shocks, and the asymptotic properties that justify treating the shift-share estimator as consistent no longer apply ([Borusyak et al., 2022](#)).

⁶If the shocks are correlated with each other, the design carries less independent identifying variation than the number of module-weeks suggests, and a shock correlated with an omitted, module-wide confound could imitate as exogenous variation in CRM or GenAI use. We estimate intraclass correlations via a hierarchical random-effects model of the shock series, following the approach in [Borusyak et al. \(2022\)](#) and the clustering diagnostics used in shift-share designs more generally ([Adão et al., 2019](#)). This route avoids defending the exogeneity of agents' module exposure shares directly, a requirement [Goldsmith-Pinkham et al. \(2020\)](#) show is demanding and one we consider implausible here, since agents' historical module reliance reflects their own workflow patterns.

⁷The measure is the Real User Monitoring (RUM) resource count for a CRM workflow module, the number of objects and network requests the module's interface retrieves to complete one render, summed over the platform's renders of that module in the week. Systems research characterises a web page by the content fetched in rendering it, principally the number of objects fetched ([Butkiewicz et al., 2011](#)), and treats a page load as a structure of linked activities whose dependencies can be recovered and traced ([Wang et al., 2013](#)). Real-user monitoring is the established means of observing a deployed system as it actually operates, in place of synthetic conditions ([Cito et al., 2015](#)), which is what allows the series to be read as realised subsystem activity.

density of the module's coded dependency structure, which its developers maintain, and the volume of work the platform routes through that module. Neither is a property of any individual agent. The share $w_{ik,t-1}$ is each module's share of agent i 's total clicks across all CRM workflow modules in week $t-1$. This ensures shares reflect the agent's prior workflow pattern and are predetermined relative to the contemporaneous shift. An agent cannot alter how many dependencies a module resolves per render, which its implementation fixes, and an agent's own renders are a small fraction of the platform's renders of any module in any week. The shift is therefore plausibly independent of any individual agent's conversion outcomes within the same week. Under the exogenous-shifts route, the orthogonality condition applies to the shift, so exposure shares do not themselves need to be exogenous (Borusyak et al., 2022). The exclusion restriction additionally requires that the instrument affect conversion only through *CRM Use*, a maintained assumption we evaluate with the diagnostics in the appendix in [Section A3.3](#).

4.3.2 Instrument for GenAI Use

GenAI Use follows the same structure, with its own shock and share. The shock is the cumulative count of qualifying OpenAI product releases assigned to a module, through a given week. The platform's generative features run on OpenAI models, so a release that expands what those models can do expands what those features can do for the agent. Unlike the CRM shock, which is measured from the platform's own operation, this one is built from OpenAI's public product-release timeline, external to the platform. Releases fall into two tiers: a core tier of GPT-3.5 and GPT-4 model releases and availability updates, and a broader interaction tier that also includes API and capability updates such as the Assistants API, Chat Completions, and file search. Each qualifying release is assigned to the individual modules in which the platform had surfaced the capability that release expands. The assignment follows the platform's own deployment, recorded module by module and summarised by family in [Table A1](#). Chat Completions releases reach the modules where outbound messages are composed, Assistants API releases the modules carrying automated sequences, file search releases the modules that retrieve documents, and core model releases the

customer-facing modules across several families. A release becomes active in the first full week after its public release date, and the shock for a module is the cumulative number of qualifying releases assigned to it through that week. The platform integrated each capability within days of its public release, so a shock dated this way never switches on before the capability was available to agents. Modules never touched by a qualifying release carry a shock of zero throughout.

Its pathway to performance runs through an agent's interaction with the affected module. An agent whose work concentrates in a module that gains new GenAI capability that week is positioned to benefit from it, independent of that agent's own tendency to use AI or their underlying skill. The share tilts the baseline click-based exposure share toward modules involving more text entry, since GenAI's productivity effects are known to be uneven across task types and concentrated in generative, writing-heavy work (Dell'Acqua et al., 2026; Noy & Zhang, 2023), and modules where an agent's engagement involves substantial typing are therefore the primary sites where GenAI capabilities are consequential. The typing-adjusted share is: $w_{ik,t-1}^{\text{type}} = \frac{w_{ik,t-1} \cdot \tau_{ik,t-1}}{\sum_k w_{ik,t-1} \cdot \tau_{ik,t-1}}$; where $\tau_{ik,t-1}$ is agent i's typing time⁴ in module k during week t-1, so modules where the agent types more receive proportionally more weight. OpenAI is an independent company operating at global scale, and its product-release timing is set by its own research and development cycle. No single agent, or the platform as a whole, is large enough to influence when OpenAI ships a new capability, and each release is made publicly available to every agent at once rather than directed at this platform specifically. OpenAI's release decisions are therefore plausibly independent of any individual agent's conversion outcomes within the same week. The exclusion restriction additionally requires that the instrument affect conversion only through *GenAI Use*, a maintained assumption we evaluate with the diagnostics in the appendix in [Section A3.3](#).

4.3.3 Instrument for Complicatedness Variables – Product and Customer

Product Complicatedness and *Customer Complicatedness* are each instrumented with an Industry-Peer Complicatedness Innovation (ICI) instrument, constructed as the unexpected component of a leave-one-agency-out industry-peer complicatedness series at the agent-week level. A focal

agency's industry peers are the other agencies that share its geography and its business model, both determined before the sample period. Geography is the agency's state; agencies in the same state operate under the same insurance regulation, carrier underwriting guidelines and market conditions¹. I. Business model classifies the agency as personal-lines, commercial-lines or mixed according to the composition of its pipeline in the pre-sample period. Industry-peer groups are held fixed throughout the sample.

For each complicatedness dimension, the industry-peer mean in each week excludes every agent in the focal agency. Dropping the whole agency removes factors that are typically shared across every agent in that agency, such as its client base, management practices or business strategy, and that could simultaneously shape the complicatedness its agents face and their performance. Following [Imbert et al. \(2022\)](#), who isolate short-run deviations of a shock series from its predictable trend, we then estimate an AR(1) projection on the industry-peer series, together with agent and week fixed effects. Each week's industry-peer complicatedness is regressed on its own value from the prior week, which captures how much of that week's level is simply a continuation of the recent trend. The residual is the ICI instrument. It is free of predictable serial dependence, fixed agent heterogeneity and shocks common to all agents, and retains only unanticipated shifts in the customer and product mix of the focal agency's industry peers, not anything specific to the focal agency or its agents. The exclusion restriction requires that the ICI instrument affect performance only through the focal agent's own realized complicatedness exposure, a maintained assumption we evaluate in [Section A3.3](#).

4.4 Addressing Endogeneity

Having an instrument for each endogenous variable, we use [Wooldridge \(2015\)](#) correlated random coefficient (CRC) model, which avoids limitations of a two-stage least squares specification.⁸ It requires only one instrument per underlying endogenous variable, and it keeps the actual value of

⁸Two-stage least squares needs a valid instrument for every endogenous regressor. Our specification has more than three. Treating each as its own endogenous regressor under 2SLS would require a separate instrument for each. Even setting that aside, plugging fitted first-stage values into a squared or interacted term does not recover the square or interaction of the true value, a problem known as the forbidden regression ([Wooldridge, 2015](#)).

each endogenous variable in the second stage, with the first-stage residual entered as an additional regressor to correct for endogeneity. Furthermore, how much an agent uses CRM or GenAI may respond to the payoff the agent expects from the technology at their own complicatedness level: an agent anticipating a larger complicatedness-contingent gain would use the technology differently. If so, the moderation slope, how much *CRM Use* or *GenAI Use* changes the complicatedness-performance relationship, is not fixed across agents but correlated with the same unobserved skill and effort already argued to bias the level terms.

The CRC model, building on [Garen \(1984\)](#), addresses this directly. An agent-specific slope cannot be estimated for each agent individually, so we instead recover its population average, and the gap between an agent's own slope and that average is proxied by the same first-stage residual used to correct the level terms. Equation 1's moderation coefficients β_5 through β_9 can therefore be written as agent-specific:

$$Y_{it} = \beta_1 \text{Complicatedness}_{it} + \beta_2 \text{Complicatedness}_{it}^2 + \beta_3 \text{GenAI Use}_{it} + \beta_4 \text{CRM Use}_{it} + \sum_k \beta_{ki} Z_{k,it} + \gamma' \mathbf{X}_{it} + \mu_i + \delta_t + \varepsilon_{it} \quad (2)$$

We recover the average slope $\bar{\beta}_k$ while correcting for this gap, giving the equation actually estimated:

$$Y_{it} = \beta_1 \text{Complicatedness}_{it} + \beta_2 \text{Complicatedness}_{it}^2 + \beta_3 \text{GenAI Use}_{it} + \beta_4 \text{CRM Use}_{it} + \sum_k \bar{\beta}_k Z_{k,it} + \varphi' \hat{\mathbf{v}}_{it} + \sum_m \varphi_m P_{m,it} + \gamma' \mathbf{X}_{it} + \mu_i + \delta_t + \varepsilon_{it} \quad (3)$$

where

$$Z_{k,it} = \{ \text{GenAI Use}_{it} \times \text{CRM Use}_{it}, \text{CRM Use}_{it} \times \text{Complicatedness}_{it}, \text{CRM Use}_{it} \times \text{Complicatedness}_{it}^2, \text{GenAI Use}_{it} \times \text{Complicatedness}_{it}, \text{GenAI Use}_{it} \times \text{Complicatedness}_{it}^2 \}, \quad k = 5, \dots, 9,$$

the five structural moderation terms of [Equation \(1\)](#), indexed from 5 because β_1 through β_4 are its

level and quadratic terms;

$$\varphi' \hat{\mathbf{v}}_{it} = \varphi_1 \hat{v}_{\text{Complicatedness},it} + \varphi_2 \hat{v}_{\text{GenAI Use},it} + \varphi_3 \hat{v}_{\text{CRM Use},it},$$

the first-stage residuals for Complicatedness, GenAI, and CRM; and

$$P_{m,it} = \{ \hat{v}_{\text{GenAI Use}} \times \text{CRM Use}, \hat{v}_{\text{GenAI Use}} \times \text{Complicatedness}, \hat{v}_{\text{CRM Use}} \times \text{GenAI Use}, \\ \hat{v}_{\text{CRM Use}} \times \text{Complicatedness}, \hat{v}_{\text{Complicatedness}} \times \text{CRM Use}, \\ \hat{v}_{\text{Complicatedness}} \times \text{GenAI Use} \}, \quad m = 4, \dots, 9,$$

each moderation term formed from two source variables, with one factor replaced by its first-stage residual.

The two coefficient blocks answer different questions. A Durbin-Wu-Hausman test on φ_1 through φ_3 asks whether Complicatedness, *GenAI Use*, and *CRM Use* are themselves endogenous, and the same test on φ_4 through φ_9 asks whether the moderation effect is separately endogenous. Both are reported at the foot of [Tables 4 to 5](#), whose columns introduce complicatedness, then technology main effects, then the moderation terms one at a time, then the full model. All specifications include agent and week fixed effects, with standard errors from a cluster(agent) bootstrap through both stages.

5 Results

[Tables 4 to 5](#) report the control-function estimates for *Customer Complicatedness* and *Product Complicatedness*, respectively, with conversion as the outcome. Column 6 is the full specification used to test H1 and H2. The two models are interpreted separately because each uses one conversion-stage complicatedness dimension as the focal endogenous portfolio measure while controlling for the other dimension.

[Insert [Table 4](#) about here]

[Insert [Table 5](#) about here]

5.1 Complicatedness and Performance

Column 1 of [Tables 4 to 5](#) establishes the baseline estimated relationship between each complicatedness dimension and conversion. *Customer Complicatedness* has a positive linear term (0.491, $p < .01$) and a negative quadratic term (-0.170, $p < .01$), giving an inverted-U relationship with conversion. Performance initially rises as *Customer Complicatedness* increases and then declines at higher levels. *Product Complicatedness* also has positive linear and negative quadratic terms (0.859, $p < .01$; -0.046, $p < .01$), but its marginal effect remains positive throughout the observed range, and conversion continues to rise at a diminishing rate across that range. The contrast is important for the theory because complicatedness alone does not imply that harmful complexity dynamics have emerged. Although the baseline is not hypothesized, we establish it before testing moderation. In the full model ([Tables 4 to 5](#), Column 6), neither *CRM Use* nor *GenAI Use* has a significant direct effect on conversion for either dimension.

5.2 Deterministic CRM Feature Use as Moderator (H1)

Column 6 of [Tables 4 to 5](#) reports the CRM interactions used to test H1 ($\text{CRM} \times \text{Customer Complicatedness} = -0.130, p < .01$; $\text{CRM} \times \text{Product Complicatedness} = -0.141, p < .01$). Figure 2A shows the fitted relationship at low and high *CRM Use*, where greater use flattens the inverted-U, lowering its peak while raising conversion at both extremes of *Customer Complicatedness*. Because the model is quadratic in complicatedness, the moderating effect is not constant. Differentiating the marginal effect of complicatedness (d) on conversion with respect to *CRM Use* (c) gives,

[Insert [Figure 2](#) about here]

$$\frac{\partial}{\partial c} \left(\frac{\partial Y}{\partial d} \right) = -0.130 + 0.118d.$$

Moderating effect of CRM use

which Figure 2B plots with 95 percent confidence bands. The moderating effect of CRM is negative and significant through the lower portion of the observed range, crosses zero at approximately 11 units, and becomes positive and significant in the upper portion ([Spiller et al., 2013](#)).

Thus, the negative moderation predicted by H1 is supported where *Customer Complicatedness* is low to moderate, but it reverses at high *Customer Complicatedness*. The reversal is informative because it arrives while the baseline customer-complicatedness relationship is still rising.

Figure 2C expresses the same moderation as the marginal return to complicatedness, drawn across the observed range of *CRM Use* at low and high *Customer Complicatedness*. Where *Customer Complicatedness* is low, the return is positive at every level of *CRM Use*, and greater use reduces it without eliminating it. Where *Customer Complicatedness* is high, the return is significantly negative when *CRM Use* is low, but higher *CRM Use* attenuates that penalty. H1 therefore holds over the lower part of the *Customer Complicatedness* range and reverses over the upper part.

Product Complicatedness follows a different pattern. Figure 4A shows the fitted relationship at low and high *CRM Use*. Conversion rises across the whole observed range at both levels, and no turning point appears. Greater *CRM Use* rotates that curve clockwise, raising conversion where *Product Complicatedness* is low and lowering it where *Product Complicatedness* is high. Repeating the derivation for *Product Complicatedness* gives a moderating effect whose variation depends on a quadratic interaction that is not significant.

[Insert *Figure 4* about here]

Figure 4B plots that effect with 95 percent confidence bands. The moderating effect of CRM is negative at every observed level of *Product Complicatedness*, and its confidence band lies entirely below zero. The moderation weakens as *Product Complicatedness* rises, but never enough to reverse it. H1 therefore holds across the whole *Product Complicatedness* range.

Figure 4C expresses the same moderation as the marginal return to complicatedness, drawn across the observed range of *CRM Use* at low and high *Product Complicatedness*. Both lines slope downward, and both stay positive at every level of *CRM Use*. Greater *CRM Use* therefore reduces the return to additional product categories.

Taken together, the CRM results support the hypothesized constraining effect of predetermined structure over most conditions in which additional complicatedness remains performance-

enhancing, while also revealing a boundary condition not specified in H1. Once *Customer Complicatedness* reaches the range in which additional complicatedness is associated with lower conversion, greater *CRM Use* attenuates that decline, where retrieval and coordination demands would otherwise be reconstructed case by case. Because interaction structure is not directly observed, this reversal is not evidence that complexity itself has emerged or been tamed. It is instead a performance pattern consistent with the possibility that predefined records and workflows become more valuable when retrieval and coordination demands begin to dominate the benefits of additional variety. The findings therefore support H1 while qualifying it. Predetermined structure reduces the marginal return to complicatedness when workers are still productively exploiting additional variety, but its value can increase once the demands associated with that variety become sufficiently consequential for performance.

5.3 Probabilistic GenAI Feature Use as Moderator (H2)

Column 6 of [Tables 4 to 5](#) reports the GenAI interactions used to test H2 ($\text{GenAI} \times \text{Customer Complicatedness} = 0.140, p < .10$; $\text{GenAI} \times \text{Product Complicatedness} = 0.271, p < .01$). Figure 3A shows the fitted relationship at low and high *GenAI Use*. The inverted-U persists at both levels, and greater *GenAI Use* raises its peak and moves that peak outward, so conversion continues to rise over a wider span of *Customer Complicatedness*. Repeating the derivation for *GenAI Use* gives a moderating effect whose variation depends on a quadratic interaction that is not significant.

[Insert Figure 3 about here]

Figure 3B plots that effect with 95 percent confidence bands. The moderating effect of GenAI is positive at every observed level of *Customer Complicatedness* and never crosses zero. The confidence band excludes zero around the sample mean and through the lower-to-middle portion of the range, but widens at the upper extreme. H2 therefore holds over the range in which the effect is estimated precisely, and the evidence thins where *Customer Complicatedness* is highest.

Figure 3C expresses the same moderation as the marginal return to complicatedness, drawn across the observed range of *GenAI Use* at low and high *Customer Complicatedness*. Both lines

slope upward. Where *Customer Complicatedness* is low, the return is positive at every level of *GenAI Use*, and greater use raises it further. Where *Customer Complicatedness* is high, the return is significantly negative when *GenAI Use* is low, and greater *GenAI Use* attenuates that penalty until it is no longer significant. H2 therefore holds at both levels of *Customer Complicatedness*.

Product Complicatedness follows the same direction with greater consistency. Figure 5A shows the fitted relationship at low and high *GenAI Use*. Conversion rises across the whole observed range at both levels, and no turning point appears. Greater *GenAI Use* rotates that curve anticlockwise, lowering conversion where *Product Complicatedness* is low and raising it where *Product Complicatedness* is high. Repeating the derivation for *Product Complicatedness* gives a moderating effect whose variation again depends on a quadratic interaction that is not significant.

[Insert *Figure 5* about here]

Figure 5B plots that effect with 95 percent confidence bands. The moderating effect of GenAI is positive at every observed level of *Product Complicatedness*, and its confidence band lies entirely above zero. The moderation weakens as *Product Complicatedness* rises, but never enough to reverse it. H2 therefore holds across the whole *Product Complicatedness* range.

Figure 5C expresses the same moderation as the marginal return to complicatedness, drawn across the observed range of *GenAI Use* at low and high *Product Complicatedness*. Both lines slope upward, and both stay positive at every level of *GenAI Use*. Greater *GenAI Use* therefore raises the return to additional product categories at both levels.

The GenAI results are consistent with the response-variety mechanism developed in H2. Probabilistic generation can supply candidate interpretations, syntheses, and responses for configurations the worker has not already encoded, and across the observed range that contribution exceeds its prompting, interpretation, and verification costs. The estimates do not directly measure response variety, however. They establish that *GenAI Use* increases the performance return to complicatedness under the identification assumptions. Response-variety augmentation remains the theory-consistent mechanism to be tested more directly in future research.

Across deterministic and probabilistic feature use, the estimated marginal effects give the

economic significance of the contrast between two interaction logics. At low *Customer Complicatedness* the marginal effect on conversion is positive, declining by 4.7 percentage points as *CRM Use* increases from low to high and rising by 6.2 points as *GenAI Use* increases. At higher *Customer Complicatedness* the marginal effect is negative, and higher *CRM Use* attenuates it by 7.0 points, while *GenAI Use* has no significant moderation. At mean *Product Complicatedness* the marginal effect declines by 5.8 points with increased *CRM Use* and rises by 9.3 points with increased *GenAI Use*, and that pattern does not change across the observed range. The findings support the hypothesized distinction between predetermined structuring and context-contingent generation while revealing that the value of structure depends on whether additional task variety is still being productively exploited.

6 Instrument Validity

The moderation results above depend on assumptions that must hold for the estimates to be interpreted causally. [Table 6](#) states each requirement, the evidence that bears on it, and where the supporting analysis appears. [Appendix A2](#) sets out the threats the design faces and [Appendix A3](#) reports the diagnostics in full. Under the assumptions the IV design maintains, the estimated moderations of portfolio complicatedness by CRM and *GenAI Use* are supported causally.

Table 6. Identification Requirements and Diagnostic Evidence

Requirement	Applies to	Test and finding	Location
Requirements of the instrumental-variables design			
Endogeneity. The regressors are correlated with unobserved determinants of conversion.	Customer complicatedness, product complicatedness, CRM Use and GenAI Use	Test: Durbin-Wu-Hausman tests on the level terms, the moderation terms and the two jointly. Finding: Exogeneity rejected in all three tests in both models.	Section 6 , Tables 4 to 5

continued on the next page

Table 6. Identification Requirements and Diagnostic Evidence (continued)

Requirement	Applies to	Test and finding	Location
Relevance. Each instrument predicts its own endogenous regressor.	All four instruments	Test: Sanderson-Windmeijer conditional F-statistics for each endogenous regressor. Finding: 10.05 to 56.98. Both complicatedness measures and GenAI Use load most strongly on their own instrument, and the two technology variables are identified jointly by the shift-share pair.	Appendix A3.1, Table A2
Exclusion restriction. Each instrument affects conversion only through its own endogenous regressor.	All four instruments	Test: Local-irrelevance diagnostics, then the exclusion test of D'Haultfœuille et al. (2024) . Finding: Local irrelevance confirmed for all four, and the exclusion test fails to reject for all four.	Appendix A3.3, Table A4
Independence. Each instrument is unrelated to unobserved determinants of conversion.	CRM Use and GenAI Use instruments	Test: Negative-control outcomes on five predetermined portfolio measures, cluster-bootstrapped. Finding: Two of five significant per instrument, the largest 0.045 standard deviations in absolute value.	Appendix A3.2, Table A3
Monotonicity. Each instrument moves its endogenous regressor in one direction across agents.	All four instruments	Design: Carried by the construction of the instruments and not separately testable.	Section 4.4
Additional requirements of the shift-share construction			
Quasi-random shock assignment. Module-week shocks are as good as randomly assigned, and exposure shares may be endogenous.	CRM Use and GenAI Use instruments	Design: Shocks set at module-week level, the module's dependency load for CRM and OpenAI release timing for GenAI (Borusyak et al., 2022 , Assumption 1).	Appendix A3.4
Many, dispersed shocks. Identification draws on many module-weeks, none carrying concentrated weight.	CRM Use and GenAI Use instruments	Test: Effective number of shocks and largest single shock weight across 1,820 module-week cells. Finding: 166.79 and 103.72 effective shocks, largest single weight 1.8 percent.	Appendix A3.4, Table A5
Mutually uncorrelated shocks. The CRM shock and the GenAI shock are uncorrelated across module-weeks, within each series and between the two, so the two instruments separate the technologies.	CRM Use and GenAI Use instruments	Test: Variance decomposition of each series across subsystem and module layers, and dependence between the two series at module-week level net of module and week effects. Finding: Within each series, correlations are moderate with no clustering that would concentrate identification. Between the two series, neither shock predicts the other, and the partial correlation is -0.012 ($p = .60$).	Appendix A3.4, Table A6

Table 4 and Table 5 (Column 6) report Durbin-Wu-Hausman tests on the level terms, on the moderation terms, and on the two jointly. Wooldridge (2015) shows that testing the significance of the control-function residual is itself a test of the null hypothesis that the corresponding regressor is exogenous, a convention applied in the information systems literature by Tiwana (2015a, 2015b). All three reject exogeneity in the full model, supporting the correlated random coefficient control-function specification. Table A8 in the appendix reports the residual terms themselves, which locate the endogeneity in complicatedness at the level and in the technology-side component of each moderation. Table 7 corroborates this with an uncorrected fixed-effects benchmark, the same regressors and interactions without the residual correction terms.

[Insert Table 7 about here]

Comparing an uncorrected specification to its corrected counterpart is standard practice for gauging the presence and direction of endogeneity bias (Ahearne et al., 2025; Angrist & Pischke, 2009; Mo & Zhang, 2025; Wooldridge, 2015). Relative to Column 6 of Table 4 and Table 5, the uncorrected estimates are smaller in magnitude but keep the same sign, and the GenAI $\times$ complicatedness interaction loses significance entirely. This pattern matches the endogeneity argument in Section 4.3. The control-function correction is therefore necessary to interpret the paper's moderation results causally, and the GenAI interaction shows most clearly what is lost without it.

7 Discussion and Conclusion

This study asks how deterministic and probabilistic IS features change the performance consequences of complicatedness in an individual knowledge worker's task portfolio. The findings support the central directional predictions of the revised theory. *CRM Use* generally reduces the marginal performance return to product and *Customer Complicatedness*, whereas *GenAI Use* increases it. The nonlinear results add an important qualification: CRM becomes relatively protective at high *Customer Complicatedness*, where the baseline marginal return to additional customer breadth has turned negative. Together, the results suggest that the performance value of IS depends

not simply on how many components a worker manages, but on how a technology changes the worker's ability to exploit or cope with the variety those components create.

The baseline relationships are important to this interpretation because they show why complicatedness should not be treated as intrinsically harmful. *Product Complicatedness* remains positively associated with conversion throughout the observed range, although at diminishing rates, and *Customer Complicatedness* initially improves conversion before reaching a turning point. This pattern is consistent with prior IS work that uses the term complexity for component variety and interconnection yet recognizes that such variety can create value ([Mocker et al., 2014](#)). Interpreted through the distinction adopted here, greater complicatedness enlarges the opportunity set available to the worker while simultaneously increasing information-processing and coordination demands. The managerial problem is therefore not necessarily to minimize complicatedness, but to preserve its useful variety while preventing the demands it creates from exceeding the worker's capacity to act effectively.

The CRM findings show why the level of analysis matters for that problem. Organizational-level research demonstrates that enterprise IS can tame complexity by changing the interaction architecture among organizational components. In multihospital systems, for example, enterprise-wide data analytics platforms replace ad hoc cross-hospital exchanges with more standardized and controlled information flows, reducing the cybersecurity consequences of system complicatedness ([Tanriverdi et al., 2025](#)). At the individual level studied here, CRM intervenes differently. Product categories and customer segments are components of a worker's task portfolio, but CRM primarily structures how the focal worker represents, retrieves, sequences, and acts on information about those components. When additional breadth remains productively exploitable, requiring heterogeneous situations to pass through predefined representations and workflows can reduce the marginal return to that breadth. When customer breadth reaches levels at which additional variety is associated with lower performance, however, the same structure becomes relatively protective. The finding does not contradict system-level complexity taming; it identifies a boundary on transferring that mechanism across levels of analysis.

The GenAI findings illuminate a complementary response to increasing complicatedness. [Nan and Tanriverdi \(2017\)](#) use requisite-variety reasoning to distinguish conditions in which additional information can select among existing responses from conditions in which the agent lacks sufficient response variety and must generate new possibilities. At the individual level, probabilistic GenAI can lower the cost of constructing context-contingent interpretations and candidate responses as product and customer configurations proliferate. Consistent with that argument, GenAI increases the marginal performance return to both dimensions of complicatedness. We do not infer that GenAI eliminates or directly 'accommodates' complexity, because the study does not observe complex interaction dynamics or directly measure the worker's response repertoire. The more defensible implication is that IS can address increasing complicatedness not only by constraining the demands placed on an actor, but also by augmenting the range of responses the actor can construct in context.

The absence of statistically significant direct effects for CRM or GenAI in the full specification reinforces the contingent nature of these mechanisms. The results do not establish that one feature class is intrinsically superior. Instead, they show that technology use changes the marginal value of task-portfolio breadth. Predetermined structure is more constraining while additional variety remains useful and becomes relatively protective when customer breadth is associated with deterioration; context-contingent generation becomes more valuable as the portfolio widens. This shifts the theoretical focus from a general question of whether IS reduces complexity to a more precise question: whether a technology primarily reduces the variety the worker must manage or increases the worker's capacity to respond to that variety.

7.1 Contributions to Research

First, the study clarifies the object of explanation in IS complexity research. Prior IS studies have often used complexity broadly for component number, variety, heterogeneity, interdependence, and the burdens associated with them. Following the sharper distinction in recent complexity-science-informed IS research, we interpret such conditions as complicatedness unless the study

theorizes or observes contingent, ad hoc, nonlinear, or emergent interaction dynamics (Benbya et al., 2020; Tanriverdi & Du, 2020; Tanriverdi et al., 2025). This reinterpretation matters substantively. Evidence that product 'complexity' can create value (Mocker et al., 2014) becomes evidence that complicatedness need not be intrinsically harmful. Our own results reinforce that point: *Product Complicatedness* remains performance-enhancing throughout the observed range, and *Customer Complicatedness* is beneficial over part of its range. Distinguishing component breadth from emergent complexity therefore changes the theoretical question from how to suppress a harmful condition to how actors can capture the value of variety while managing the demands it creates.

Second, the study identifies level of analysis as a boundary condition on how IS manages complicatedness and potential complexity. In organizational systems, an enterprise technology can directly restructure exchanges among constituent units, as in the cross-hospital interaction architecture studied by Tanriverdi et al. (2025). In an individual task portfolio, by contrast, the focal worker mediates dependencies among products, customers, activities, and prior interactions. Deterministic IS therefore acts primarily through the worker's representations and workflows rather than by redesigning autonomous interunit exchanges. The CRM results show why this distinction is consequential: more predetermined structure does not automatically become more valuable as the individual's component set grows. Its marginal value depends on whether the worker is still extracting value from additional breadth or has entered a region in which the associated demands are reducing performance. This provides a theoretically grounded boundary on cross-level extrapolation of complexity-taming mechanisms.

Third, the study distinguishes two complementary technological responses to increasing task variety: structuring demands and augmenting response variety. Deterministic CRM features address complicatedness by imposing predefined representations, states, and workflows that can reduce retrieval and coordination demands but can also constrain heterogeneous situations. Probabilistic GenAI features can instead generate context-contingent candidate responses, potentially expanding the repertoire available to the worker as task variety increases. This distinction connects IS complexity research with requisite-variety reasoning (Ashby, 1956; Nan & Tanriverdi, 2017) without

equating *GenAI Use* with response variety itself. Empirically, the opposite moderation patterns for CRM and GenAI are consistent with these mechanisms. The contribution is therefore not that response-capacity augmentation is wholly new - related ideas appear in [Mocker et al. \(2014\)](#) and [Nan and Tanriverdi \(2017\)](#) - but that we theorize and estimate how deterministic structuring and probabilistic generation have different marginal performance consequences as individual-level task portfolio complicatedness rises.

7.2 Contributions to Practice

For practice, the results caution against treating either greater structure or greater generative capability as a universal response to complicated work. Managers should first diagnose whether portfolio breadth is creating useful variety that workers can still exploit or whether retrieval, coordination, and continuity demands are beginning to dominate. Deterministic CRM features can provide a valuable structural backbone, but adding or intensifying predefined workflow requirements may reduce the return to breadth when workers still benefit from heterogeneous products and customers. Probabilistic GenAI becomes more valuable as breadth rises in our setting, but the mechanism still requires worker judgment, verification, and governance. A practical design implication is therefore to preserve reliable records and workflow state while deploying generative support where context-specific synthesis and response construction are likely to expand workers' effective capacity.

7.3 Limitations and Future Research

Several limitations delimit these conclusions and create opportunities for further research. First, the study directly measures portfolio complicatedness, feature-use intensity, and conversion performance, but not complexity itself. We do not observe whether interactions among products, customers, activities, and prior actions become structured, ad hoc, linear, nonlinear, or emergent. Future studies could measure these dynamics using workflow sequences, conversational traces, process-mining measures of branching and iteration, or direct indicators of cross-component de-

pendence. Second, *CRM* and *GenAI Use* are behavioral intensity measures rather than direct measures of the mechanisms proposed here. Future work should test whether *CRM Use* actually reduces interaction variety or representational ambiguity and whether *GenAI* expands workers' effective response repertoires, as well as when verification burden offsets that expansion. Third, the empirical setting concerns individual insurance agents pursuing a desired conversion outcome. Replication in other knowledge-work domains is needed to establish how task objectives, expertise, regulation, error costs, and the autonomy of system components alter the relative value of structuring demands versus augmenting response variety. Finally, the nonlinear customer result deserves direct investigation: future research should examine whether the point at which predetermined structure becomes protective corresponds to measurable changes in coordination burden, interaction branching, or worker capacity rather than treating a performance turning point as evidence that complexity has emerged.

7.4 Conclusion

Complicated knowledge work is not simply a problem of having too many components. Product and customer breadth can create opportunities as well as demands, and the value of IS depends on how technology changes the worker's capacity to realize those opportunities. In this study, deterministic *CRM Use* generally reduces the marginal performance return to complicatedness while becoming relatively protective when customer breadth reaches a performance-reducing region; probabilistic *GenAI Use* increases the marginal return to both product and *Customer Complicatedness*. These findings suggest a level-sensitive account of IS and complexity: organizational systems may benefit when technology restructures interactions among units, whereas individual knowledge workers may also benefit when technology expands the repertoire of responses available for heterogeneous situations. By separating complicatedness from complexity and predetermined structuring from response-variety augmentation, the study provides a more precise foundation for explaining when different forms of IS help workers perform in increasingly complicated task environments.

Figure 1. Customer Acquisition Pipeline

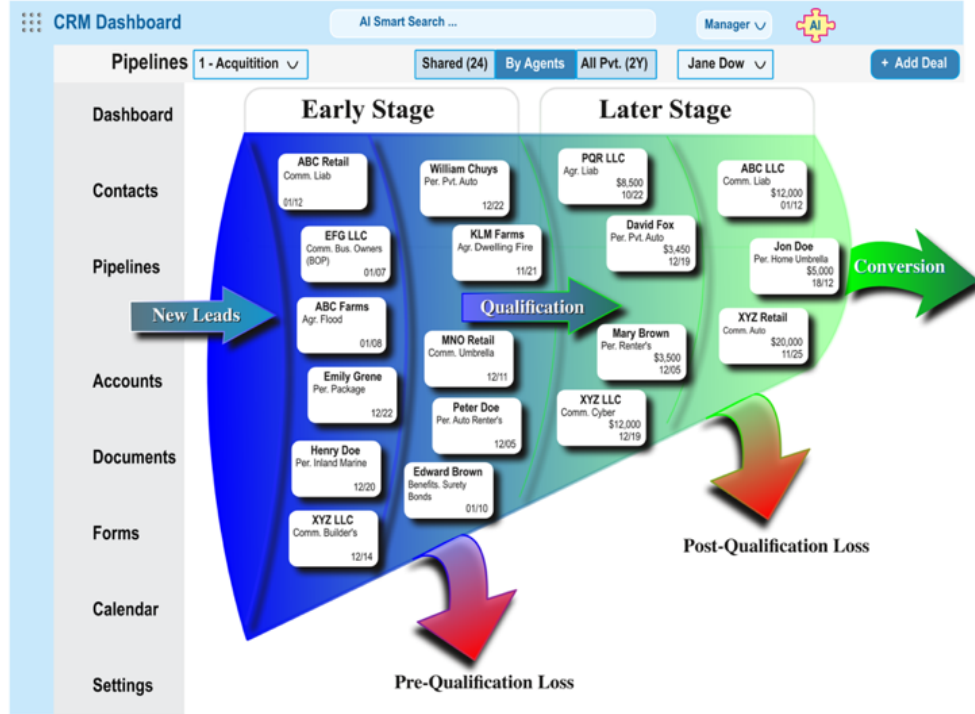


Table 2. Descriptive Statistics

	Mean (1)	SD (2)	Min (3)	Max (4)
Conversion rate [†]	11.45	22.84	0	83.33
Product complicatedness [†] (qualification-stage)	10.31	8.02	0	44.00
Product complicatedness [†] (conversion-stage)	4.97	7.36	0	42.86
Customer complicatedness [†] (qualification-stage)	9.97	5.33	0	26.67
Customer complicatedness [†] (conversion-stage)	4.36	5.86	0	26.67
Process activity burden [†]	2.44	4.23	0	54.00
GenAI Use [†]	4.83	31.73	0	216.83
CRM Use [†]	425.40	619.53	0	2627.24
Portfolio value [†]	25.63	9.48	0	35.86
Portfolio age	5.30	2.22	0	7.00
Bundling [†]	6.53	2.76	0	24.00
Portfolio size	41.66	59.90	0	231.00

Notes: N = 16,408 for all variables. [†] Reported after applying confidential linear scaling transformations.¹

Table 3. Correlations

# Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
1 Conversion rate [†]	1.00											
2 Product complicatedness [†] (qualification-stage)	-0.08*	1.00										
3 Product complicatedness [†] (conversion-stage)	0.21*	0.42*	1.00									
4 Customer complicatedness [†] (qualification-stage)	-0.09*	0.43*	0.15*	1.00								
5 Customer complicatedness [†] (conversion-stage)	0.04*	0.24*	0.32*	0.59*	1.00							
6 Process activity burden [†]	0.42*	0.00	0.21*	0.11*	0.20*	1.00						
7 GenAI Use [†]	0.06*	-0.04*	0.01	-0.03*	-0.01	0.02	1.00					
8 CRM Use [†]	-0.03*	0.06*	0.07*	0.19*	0.20*	0.11*	0.18*	1.00				
9 Portfolio value [†]	0.07*	0.22*	0.10*	0.20*	0.13*	0.07*	-0.00	-0.12*	1.00			
10 Portfolio age	-0.41*	0.19*	-0.13*	0.03*	-0.10*	-0.35*	0.00	-0.16*	-0.01	1.00		
11 Bundling [†]	0.03*	0.37*	0.46*	0.11*	0.11*	0.10*	-0.05*	-0.05*	0.31*	-0.20*	1.00	
12 Portfolio size	-0.30*	0.14*	0.11*	0.29*	0.26*	-0.02	-0.04*	0.37*	-0.08*	-0.15*	-0.02	1.00

Notes: * Significance at $p < 0.01$. [†] Reported after applying confidential linear scaling transformations.

Table 4. Results for Customer Complicatedness

	Conversion					
	(1)	(2)	(3)	(4)	(5)	(6)
Customer Complicatedness	0.491*** (0.046)	0.491*** (0.068)	0.491*** (0.068)	0.522*** (0.065)	0.513*** (0.071)	0.554*** (0.067)
Customer Complicatedness ²	-0.170*** (0.025)	-0.166*** (0.025)	-0.167*** (0.025)	-0.177*** (0.028)	-0.161*** (0.027)	-0.173*** (0.030)
GenAI Use		-1.397 (1.412)	-1.462 (1.408)	-1.248 (1.338)	-1.512 (1.430)	-1.246 (1.343)
CRM Use		0.058 (0.146)	0.074 (0.144)	-0.005 (0.139)	0.072 (0.146)	-0.009 (0.140)
Structural moderation (CRC) terms						
GenAI Use × CRM Use			0.044 (0.069)	0.059 (0.068)	0.036 (0.071)	0.048 (0.068)
CRM Use × Customer Complicatedness				-0.116*** (0.029)		-0.130*** (0.030)
CRM Use × Customer Complicatedness ²				0.060*** (0.019)		0.059*** (0.019)
GenAI Use × Customer Complicatedness					0.096 (0.068)	0.140* (0.074)
GenAI Use × Customer Complicatedness ²					0.022 (0.034)	0.009 (0.033)
Product Complicatedness	0.235*** (0.047)	0.235*** (0.066)	0.235*** (0.067)	0.245*** (0.064)	0.241*** (0.066)	0.255*** (0.062)
Process activity burden	0.243*** (0.014)	0.256*** (0.021)	0.257*** (0.021)	0.255*** (0.020)	0.257*** (0.021)	0.256*** (0.019)
Product Complicatedness (qualification-stage)	0.034 (0.033)	0.029 (0.045)	0.028 (0.046)	0.035 (0.045)	0.027 (0.046)	0.035 (0.044)
Customer Complicatedness (qualification-stage)	0.033 (0.048)	0.039 (0.060)	0.040 (0.060)	0.028 (0.057)	0.040 (0.060)	0.027 (0.057)
Portfolio value	0.213*** (0.029)	0.212*** (0.037)	0.212*** (0.037)	0.206*** (0.036)	0.212*** (0.037)	0.206*** (0.036)
Portfolio age	-0.446*** (0.030)	-0.449*** (0.035)	-0.450*** (0.035)	-0.445*** (0.035)	-0.449*** (0.035)	-0.443*** (0.034)
Bundling	0.151*** (0.036)	0.156*** (0.042)	0.157*** (0.042)	0.147*** (0.041)	0.157*** (0.043)	0.146*** (0.041)
Portfolio size	0.381*** (0.046)	0.363*** (0.058)	0.362*** (0.058)	0.357*** (0.057)	0.362*** (0.059)	0.355*** (0.057)
Constant	0.895*** (0.049)	0.475 (0.440)	0.453 (0.439)	0.543 (0.417)	0.437 (0.446)	0.543 (0.419)
N	16408	16408	16408	16408	16408	16408
Adj. R ²	0.680	0.682	0.682	0.684	0.682	0.684
Overall F	239.78	180.96	151.22	126.43	125.63	109.20
Endogeneity correction terms	Yes	Yes	Yes	Yes	Yes	Yes
DWH test: Level Terms	33.69***	21.87***	20.72***	18.27***	21.17***	18.19***
DWH test: Moderation Terms	—	—	0.05	1.29	0.95	2.19**
DWH test: Joint	33.69***	21.87***	13.15***	9.77***	10.19***	8.72***

Notes: Bootstrap standard errors in parentheses for coefficient rows. Stars: * p<.10, ** p<.05, *** p<.01. Agent and week fixed effects included.

Table 5. Results for Product Complicatedness

	Conversion					
	(1)	(2)	(3)	(4)	(5)	(6)
Product Complicatedness	0.859*** (0.067)	0.871*** (0.086)	0.873*** (0.087)	0.903*** (0.085)	0.934*** (0.092)	0.985*** (0.091)
Product Complicatedness ²	-0.046*** (0.017)	-0.045*** (0.017)	-0.045*** (0.017)	-0.042** (0.017)	-0.050*** (0.017)	-0.048*** (0.018)
GenAI Use		-1.371 (1.443)	-1.442 (1.445)	-1.202 (1.368)	-1.476 (1.467)	-1.198 (1.375)
CRM Use		0.053 (0.148)	0.071 (0.146)	0.045 (0.140)	0.071 (0.149)	0.040 (0.141)
Structural moderation (CRC) terms						
GenAI Use × CRM Use			0.050 (0.070)	0.067 (0.068)	0.040 (0.068)	0.049 (0.062)
CRM Use × Product Complicatedness				-0.121*** (0.035)		-0.141*** (0.035)
CRM Use × Product Complicatedness ²				0.007 (0.013)		0.008 (0.013)
GenAI Use × Product Complicatedness					0.189** (0.089)	0.271*** (0.094)
GenAI Use × Product Complicatedness ²					-0.014 (0.013)	-0.014 (0.014)
Customer Complicatedness	-0.098** (0.047)	-0.111* (0.059)	-0.112* (0.059)	-0.097* (0.057)	-0.114* (0.059)	-0.096* (0.056)
Process activity burden	0.231*** (0.015)	0.244*** (0.021)	0.244*** (0.021)	0.244*** (0.020)	0.245*** (0.021)	0.245*** (0.020)
Product Complicatedness (qualification-stage)	-0.016 (0.034)	-0.023 (0.045)	-0.023 (0.045)	-0.011 (0.044)	-0.025 (0.046)	-0.010 (0.043)
Customer Complicatedness (qualification-stage)	0.056 (0.048)	0.065 (0.060)	0.065 (0.061)	0.050 (0.057)	0.066 (0.061)	0.050 (0.058)
Portfolio value	0.221*** (0.029)	0.221*** (0.037)	0.221*** (0.037)	0.214*** (0.035)	0.221*** (0.037)	0.214*** (0.035)
Portfolio age	-0.422*** (0.029)	-0.425*** (0.035)	-0.425*** (0.036)	-0.420*** (0.034)	-0.426*** (0.036)	-0.420*** (0.034)
Bundling	0.126*** (0.036)	0.130*** (0.042)	0.131*** (0.042)	0.117*** (0.040)	0.132*** (0.042)	0.117*** (0.041)
Portfolio size	0.363*** (0.046)	0.347*** (0.059)	0.347*** (0.059)	0.338*** (0.057)	0.346*** (0.060)	0.336*** (0.057)
Constant	0.783*** (0.042)	0.374 (0.448)	0.350 (0.448)	0.437 (0.425)	0.338 (0.455)	0.439 (0.428)
N	16408	16408	16408	16408	16408	16408
Adj. R ²	0.676	0.678	0.678	0.680	0.678	0.681
Overall F	235.86	178.61	149.58	126.13	124.16	108.54
Endogeneity correction terms	Yes	Yes	Yes	Yes	Yes	Yes
DWH test: Level Terms	47.03***	25.83***	24.74***	19.99***	25.13***	19.53***
DWH test: Moderation Terms	—	—	0.07	1.78	1.62	3.33***
DWH test: Joint	47.03***	25.83***	15.57***	10.90***	12.32***	10.27***

Notes: Bootstrap standard errors in parentheses for coefficient rows. Stars: * p<.10, ** p<.05, *** p<.01. Agent and week fixed effects included.

Table 7. Uncorrected Fixed-Effects (No Control-Function Correction) Full Model Results

	Conversion			
	Customer complicatedness (FE-OLS)		Product complicatedness (FE-OLS)	
	(1)		(2)	
Complicatedness	0.332***	(0.049)	0.601***	(0.065)
Complicatedness ²	-0.181***	(0.035)	-0.048**	(0.020)
GenAI Use	0.024	(0.048)	0.053	(0.039)
CRM Use	-0.050*	(0.029)	0.001	(0.024)
GenAI Use × CRM Use	-0.014	(0.041)	-0.005	(0.039)
CRM Use × Complicatedness	-0.095***	(0.023)	-0.088***	(0.025)
CRM Use × Complicatedness ²	0.055***	(0.020)	0.002	(0.013)
GenAI Use × Complicatedness	0.026	(0.030)	0.042	(0.028)
GenAI Use × Complicatedness ²	0.014	(0.036)	-0.020	(0.012)
Product Complicatedness	0.420***	(0.037)		
Customer Complicatedness			0.093**	(0.035)
Process activity burden	0.241***	(0.021)	0.238***	(0.021)
Product Complicatedness (qualification-stage)	0.028	(0.035)	0.029	(0.036)
Customer Complicatedness (qualification-stage)	0.038	(0.048)	0.028	(0.048)
Portfolio value	0.209***	(0.030)	0.217***	(0.030)
Portfolio age	-0.439***	(0.033)	-0.439***	(0.033)
Bundling	0.131***	(0.038)	0.128***	(0.038)
Portfolio size	0.379***	(0.048)	0.384***	(0.049)
Constant	0.934***	(0.036)	0.802***	(0.023)
N	16,408		16,408	
Agent/Time FEs	Yes/Yes		Yes/Yes	
R-squared	0.693		0.689	
Adj. R-squared	0.681		0.676	
Overall F	78.72		86.59	
Mean VIF (uncentered)	3.555		3.545	

Notes: Standard errors clustered two-way by agent and week in parentheses. Compare to the column 6 of [Table 4](#) (*Customer Complicatedness*) and [Table 5](#) (*Product Complicatedness*) for the corresponding control-function-corrected estimates. *** p<0.01, ** p<0.05, * p<0.1.

Figure 2. Deterministic CRM Feature Use as a Moderator of Customer Complicatedness

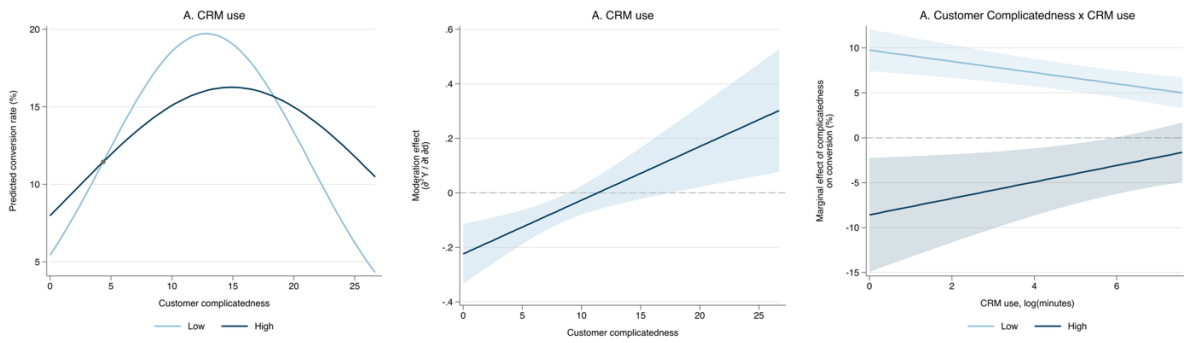


Figure 3. Probabilistic GenAI Feature Use as a Moderator of Customer Complicatedness

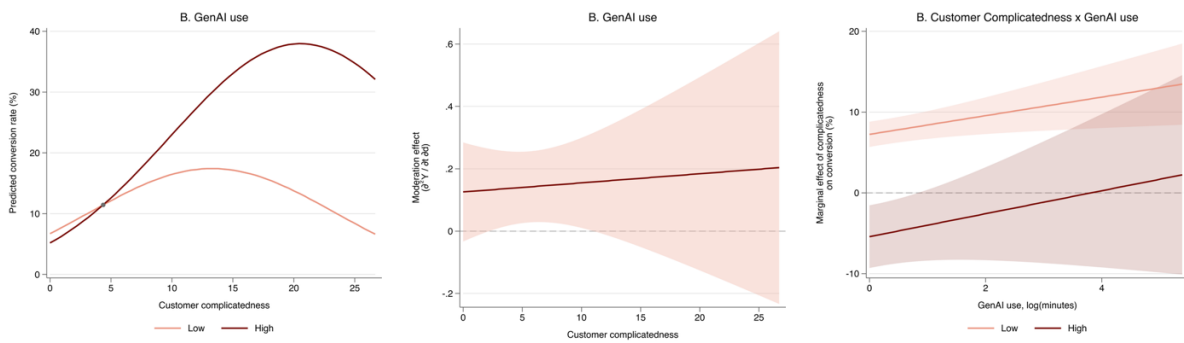


Figure 4. Deterministic CRM Feature Use as a Moderator of Product Complicatedness

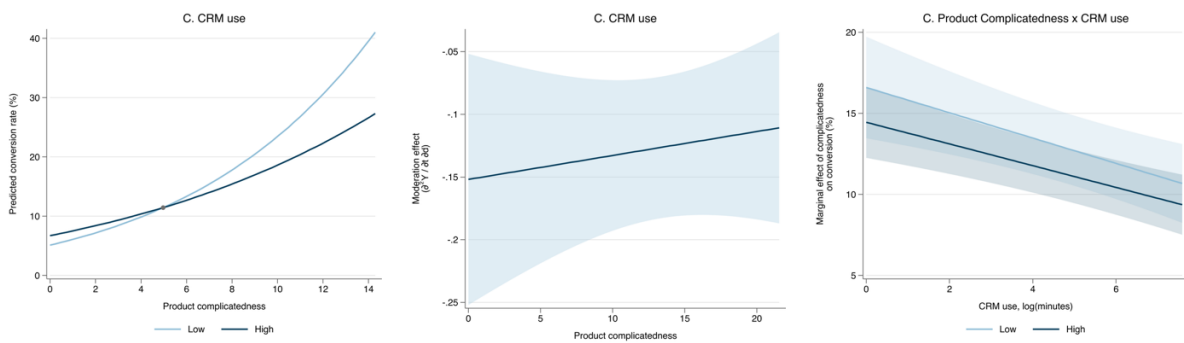
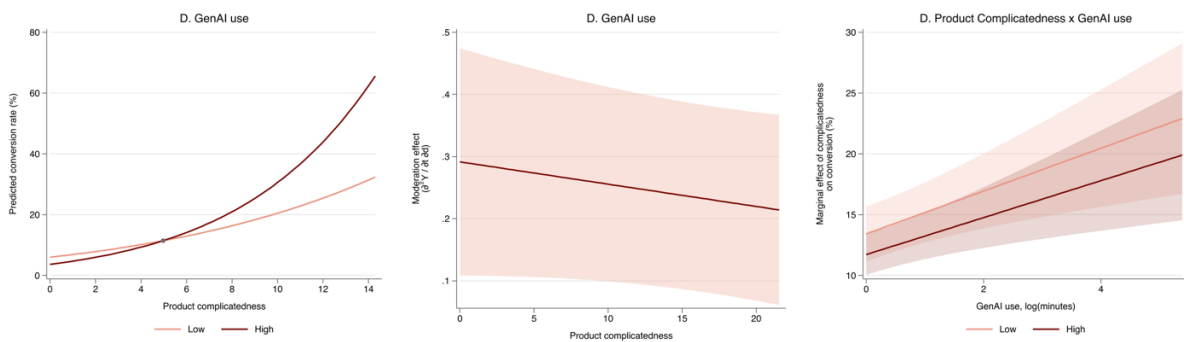


Figure 5. Probabilistic GenAI Feature Use as a Moderator of Product Complicatedness



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Appendix

A1 Platform Module Functionality

Table A1 describes the functional module groups of the study platform, defined by the platform's own architecture and used to construct the technology-use measures. The ten groups condense the platform's 65 modules, and exposure shares and shocks are constructed at the module level. A deterministic feature returns the same output given the same inputs and configuration. Interactions with these features constitute *CRM Use*. A probabilistic feature produces model-generated output that may vary across identical invocations and that the agent accepts, edits or reverts before use. These interactions are excluded from *CRM Use* and counted toward *GenAI Use*. A small number of probabilistic facilities surface in several module groups, wherever text is composed, transcribed or summarised, and the modules in which each facility appears are set by the platform's own deployment, shown by group in the table. Because one facility can appear in several modules, each invocation is classified at the level of the individual logged action and not inferred from the module.

Table A1. Platform Module Groups and Feature Classification

#	Module group	Deterministic features	Probabilistic features
G1	Opportunity management	Agents create deal records and move them through the stages of a pipeline. Records are assigned to agents, filtered by stage or date, and updated in bulk. Work built here over the period added filters, quick views, checklists, revenue fields and bulk actions. Sequences of actions run automatically when a deal reaches a stage or a date passes.	Inside a deal record the tool drafts the message an agent is about to send, and can pull the record's activity into a short summary.
G2	Customer and contact records	The account and contact records sit at the centre of the platform. Agents create records, merge duplicates, link contacts to the accounts they belong to, and search the full history of a record. Work built here over the period improved merging and duplicate handling, added search across the record timeline, and gave agencies control over which fields are required when a record is created.	The tool writes and reshapes notes on a record. It also derives the industry classification code for a commercial account from the details the agent has entered.
G3	Policy and coverage records	Policy and coverage records are held here, along with the carrier reference data they depend on. Agents renew policies, cancel them, record premium transactions and attach claim information. Work built here over the period added policy custom fields, carrier and category management, and two way synchronisation with the agency management systems the agency already runs.	The tool supplies the industry classification that describes a commercial account for coverage purposes, and can condense policy activity into a readable summary.

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Table A1. Platform Module Groups and Feature Classification (continued)

#	Module group	Deterministic features	Probabilistic features
G4	Campaign and list management	Agents send marketing to groups of customers rather than to one person. They build recipient lists, manage templates, schedule sends and handle unsubscribes. Work built here over the period added list filters, template sharing, campaign analytics and synchronisation of campaign activity with the agency management system.	For campaign content the tool produces a first draft, reworks the tone or the length, and renders the same message in another language.
G5	Individual communication	This is where an agent talks to one customer at a time, by email, text or phone. Messages are routed to the right record, scheduled, and kept as a history the agent can read later. Work built here over the period added text scheduling and editing, incoming message mapping, call history filters and call forwarding.	The tool composes replies to incoming messages. For phone calls it turns the recording into text and reduces the conversation to a summary.
G6	Customer feedback	The agency collects structured feedback from customers and routes satisfied customers to public review sites. Responses are captured, filtered and reported. Work built here over the period added feedback filters across the platform, notifications to the responsible agent, and reporting on responses. The score itself is entered by the customer.	After a customer leaves feedback or a public review, the tool composes the agency's reply.
G7	Workflow automation	Agencies build sequences of actions that run on their own once a condition is met. A sequence can send an email, send a text, create a task, set a reminder or create a deal. Triggers include renewal dates, cross sell conditions and record states. Work built here over the period added tracking of running sequences, analytics on what was sent, holiday pauses and the ability to let named agents bypass a sequence.	The tool writes the message content carried by a sequence, and the notes recorded as the sequence runs.
G8	Task and schedule coordination	Tasks and reminders keep individual and team work on schedule. Agents create tasks, assign them, work from templates and link them to the record the work concerns. Work built here over the period added filters by completion date and template, bulk assignment, past due notifications and a scheduling page.	The tool writes task notes and condenses what has happened on the record a task belongs to.
G9	Document and signature management	Documents are created, stored, retrieved and signed here, and agencies build their own forms. Work built here over the period added signature templates, sharing several documents in one request, copying additional recipients, editing documents still open for signature and an archive of completed ones.	The tool produces and revises the wording inside documents and forms.

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Table A1. Platform Module Groups and Feature Classification (continued)

#	Module group	Deterministic features	Probabilistic features
G10	Reporting and administration	This is where the agency configures the platform and reads reports on it. Administrators set access, control which fields are required, set the fiscal year and goals, and configure the automation rules used elsewhere. Reporting aggregates what has been recorded, such as the book of business by carrier or policy status. Work built here over the period added report grouping and filters, multi factor authentication and finer restrictions on what agents can change.	The tool writes the wording of templates configured here, and condenses recorded activity into summaries for review.

A2 Threats to Identification

This section sets out the threats to identification the design faces, the reason each arises in this setting, the evidence that bears on it, and the risk that remains. Each threat is stated in the form that would invalidate the estimate, so the evidence can be read against a specific violation. The threats are grouped by the object they act on. The first group concerns the shift-share instruments for *CRM Use* and *GenAI Use*. The second concerns the leave-one-agency-out industry-peer instrument for product and *Customer Complicatedness*. The third concerns how the variables are constructed. Where a threat is examined empirically, the paragraph points to the diagnostic in [Section A3](#).

A2.1 Threats to the CRM Use and GenAI Use Instruments

Exposure endogeneity. Each shift-share instrument combines a module-week shock with an agent's prior reliance on each module. An agent's historical reliance reflects role, skill, portfolio composition and habit, all of which bear on performance. Under the exogenous-shifts route the identifying assumption is placed on the shocks, and the exposure shares are permitted to be endogenous ([Borusyak et al., 2022](#)). The quasi-experimental content comes from the combination. The share is measured in week $t - 1$ and is settled before the week's module-specific movement occurs, so exposure could only have been adjusted in anticipation of a shock that had yet to happen. [Section A3.6](#) adds a direct check for the GenAI instrument. Placebo shocks with the same structure as the realised shock, but randomly paired with modules and weeks, produce no first stage on average, so the first stage does not arise from the exposure shares alone.

Attribution of a shock to an individual agent. If one agent's activity moved a module-week shock, the shock would carry that agent's own circumstances into the instrument. The CRM shock is summed over every render of the module on the platform, so its level is set by the module's coded dependency structure, which its developers maintain, and by the renders of the platform's whole user base. [Table A4](#) reports an effective sample of 166.79 module-weeks for the CRM shock and

103.72 for the GenAI shock, with the largest single module-week carrying 1.3 and 1.8 percent of the identifying weight. A shock concentrated enough to reflect one agent's activity would show the opposite dispersion.

Channels reaching performance outside CRM Use. The exclusion restriction requires the CRM instrument to reach conversion only through *CRM Use*. A module's dependency load rises with the number of times the platform assembles its interface, so the relevant alternative channel is a concentration of work routed through a module that also changes conversion directly, such as a seasonal concentration of renewals in policy modules. Three features of the design bear on this channel. Week fixed effects absorb any movement common to all modules. A concentration confined to a functional subsystem would appear as a shared component in the shock series, and [Table A5](#) shows that the module-specific component carries at least as much of the variance as the module-group component. The negative-control outcomes in [Section A3.2](#) test whether the CRM instrument predicts portfolio conditions it should not, and the exclusion-restriction test in [Section A3.3](#) does so without naming a channel in advance. Changes in how long an action takes are recorded within *CRM Use*, which is measured as time actively engaged with the platform, and so do not form a separate channel.

Correlation across modules within a shock series. A seasonal or business-cycle pattern shared across related modules would make the shocks move in a small number of coordinated clusters, which would undermine the many-shocks condition ([Borusyak et al., 2022, 2025](#)). [Table A5](#) decomposes the variance of the module-week shock into a module-group component, a module-level component and a residual. For the CRM shock the module-level component is 0.281 (SE 0.086), the module-group component 0.249 (SE 0.176) and the residual 0.046 (SE 0.044). For the GenAI shock the module-level component is 0.410 (SE 0.069), the module-group component 0.373 (SE 0.085) and the residual 0.199 (SE 0.030). Identifying variation is therefore distributed across modules within groups and is not concentrated in a few coordinated clusters.

Bundling of the GenAI shock with concurrent platform work. The GenAI shock counts qualifying OpenAI releases and nothing else, so platform work on deterministic features contributes to no part of the series and the two shocks never mix events. What remains is a question of timing, that a week carrying an OpenAI-dated shock could also carry platform changes to deterministic features. Three properties of the design bear on this. Each logged action is classified at logging, so a minute recorded as GenAI interaction is never recorded as CRM interaction, and a release touching deterministic features has no mechanical route into the GenAI-use measure. Identification comes from the differential pattern of exposure across many module-week cells over the whole window, so cells carrying one kind of release alone continue to contribute independent variation. *CRM Use* is separately instrumented in the same model, so performance moving through a concurrent change to deterministic features is attributed to that variable. Whether the two series in fact move together

is taken up next.

Correlation between the two shock series. If the CRM and GenAI shocks moved together, the two instruments would carry the same information. Correlating the two series at the module-week level, net of what is common to each module and each week, gives $r = -0.012$ ($p = .60$). A regression with module and week fixed effects and standard errors clustered by module gives a coefficient of -0.014 (SE 0.052 , $p = .784$) for the GenAI shock on the CRM shock, and -0.011 (SE 0.040 , $p = .786$) with the two reversed. The two shock series therefore carry separate information. The two technology-use variables are identified jointly by the pair of instruments (Table A2).

Thin identifying variation. The observation window contains few release waves, which raises the question whether the shift-share asymptotics apply. Borusyak et al. (2022) state the requirement as many uncorrelated shocks with a small concentration of exposure weights, and report that their approximation performs well at a concentration corresponding to roughly twenty effective shocks. The count that governs this is module-week cells, because each release is mapped to the modules whose capability it touches. Table A4 reports 103.72 effective module-week shocks for the GenAI series and 166.79 for the CRM series. Because the GenAI shock is a cumulative count, its module-week cells are not independent of one another, and the assignment of releases to modules and weeks was realised once. Section A3.6 therefore reports randomization inference at the level at which the shock was assigned, alongside the exposure-robust standard errors in Table A6.

Use of the same models outside the platform. The GenAI shock is dated by OpenAI's public release announcements, so a release that reaches this platform's agents also reaches everyone else, including through their own use of these models away from the platform. If such use moved conversion, the instrument could reach performance without passing through *GenAI Use* on the platform. Any effect common to a release week is absorbed by the week fixed effects, since every agent meets the same announcement at the same time. What identifies the instrument is differential exposure across modules within a week, so the channel would require agents whose prior work concentrated in the modules a release touched to benefit differentially from their own outside use in that same week. That is a substantially narrower claim than the common release effect the week fixed effects already remove. Section A3.6 bears on this. If *GenAI Use* responded to the public availability of the models and not to what the platform deployed, a placebo schedule drawing on the same releases and activation weeks would reproduce the first stage. It does not.

A2.2 Threats to the Product and Customer Complicatedness Instrument

Predictable movement in peer complicatedness. The industry-peer complicatedness series carries a predictable trend, which would correlate with the focal agent's own lagged portfolio state. The instrument is the residual from projecting the series on its own prior-week value together with agent and week fixed effects, following Imbert et al. (2022). What survives is the unanticipated

week-to-week movement in the portfolio mix of the focal agency’s industry peers.

Shocks shared within an industry-peer group. Week fixed effects absorb what is common to every agent. Because industry peers share a state and a business model, the demanding case is a shock specific to one state and line of business, such as a change in carrier underwriting guidelines or a regional shift in demand, that moves peers’ complicatedness and the focal agent’s conversion in the same week. Where such a shock raises peer complicatedness and the focal agent responds by widening the portfolio, the effect runs through the instrumented variable, which is the intended channel. The threat is a shock that moves commercial conditions while leaving the focal portfolio’s composition unchanged. Several such channels are captured by controls in the specification. Deal volume is captured by portfolio size (Sabnis et al., 2013). Promotional and bundled offers are captured by bundling intensity (Balachander et al., 2010; Schmitz et al., 2014). Price and value movements are captured by portfolio value. Deal-cycle velocity is captured by portfolio age (Guesalaga & Kapelianis, 2015). Underwriting and approval friction appears in process activity burden, which counts approval requests, documentation and coordination steps (Aral et al., 2012; Verbeke et al., 2011). These controls are measured in the same week as the outcome, and Section A3.8 shows that the moderation results are unchanged when they are removed or replaced with their lagged values. The negative-control outcomes in Section A3.2 test whether the instrument predicts lagged values of these same portfolio measures.

Agency-level resource reallocation. A further channel is an agency shifting resources among its agents in response to movements among its industry peers, which would reach conversion while leaving the agent’s own portfolio composition unchanged. Jointly handled deals are excluded from the sample, which removes the team-based form of this channel. Agents remain within one agency across the window, so agent fixed effects absorb any time-invariant agency allocation pattern. What remains is reallocation that varies within a week.

Channels we have not anticipated. The arguments above name the channels we can foresee. An unnamed channel can still be tested for, because where a change in the instrument leaves the endogenous variable unmoved, any remaining effect on performance must travel by some other route, whatever that route is. The test of D’Haultfœuille et al. (2024) in Section A3.3 turns on this, and it fails to reject for customer and *Product Complicatedness* alongside the two technology-use variables.

A2.3 Threats from Measurement

Contamination between the two CRM and GenAI Use measures. Both feature classes reside on one platform, which raises the possibility that minutes recorded as *CRM Use* include time spent invoking and reading generated output. The two measures are separated at construction. The underlying record is the individual logged action, and each action carries the module it belongs to and whether

it was an interaction with the generative tool or with a deterministic feature. Aggregation to the agent-week follows that classification, so an action counted toward *GenAI Use* is never added into *CRM Use*.

Mechanical correlation between the count and portfolio size. Complicatedness counts distinct categories across the deals an agent holds, and conversion is computed over that same set of deals. An agent holding more deals is likely to span more categories, so the count rises with portfolio size by arithmetic, and it rises at a decreasing rate because each additional deal is less likely to introduce a category the agent does not already hold. Portfolio size is controlled for alongside portfolio age, bundling intensity and portfolio value, but an additive control does not remove the exposure of the quadratic term to how the count is constructed. [Section A3.7](#) reports a permutation test that addresses this directly.

Potentially post-treatment controls. Process activity burden counts transactions and activities recorded in the CRM, so it may be produced by *CRM Use*, which would make it a mediator and the estimate on *CRM Use* a direct effect. Portfolio age and portfolio value are predicted by the instruments at some lags ([Section A3.2](#)). Conditioning on a consequence of the treatment or of the instrument can bias the terms it is meant to leave alone. [Section A3.8](#) re-estimates the full specification without these controls and with their lagged values.

A3 Identification Diagnostics

A3.1 First-Stage Instrument Relevance

A weak instrument could make the causal estimates unreliable, so we first confirm that the excluded instruments predict each endogenous variable. [Table A2](#) reports first-stage results. Each model instruments three source endogenous variables, the focal complicatedness measure, *CRM Use* and *GenAI Use*, and the table displays all four distinct variables appearing across the two models. Both complicatedness measures and *GenAI Use* are predicted most strongly by their own instrument. *CRM Use* loads on both shift-share instruments, so the two technology variables are identified jointly by that pair. Sanderson-Windmeijer conditional F-statistics, constructed for this case, range from 10.05 to 56.98. The complicatedness first stages are strong, and the two technology first stages lie close to the conventional threshold of 10.

A3.2 Negative-Control-Outcome Falsification Tests

The instruments could fail the exclusion restriction if an unobserved factor connected them to performance through a channel other than the endogenous variables. [Danieli et al. \(2026\)](#) formalize this as an alternative-path outcome, an unobserved threat with an observable proxy, a negative-

control outcome, that the instrument should not predict if the exclusion restriction holds. We use five portfolio measures as negative controls, portfolio size, portfolio value, portfolio age, bundling and process activity burden. Each is measured before the week in which the instrument is realised, at lags 1 to 4 for the two complicatedness instruments, and at lags 2 to 4 for the two shift-share instruments, whose exposure shares are themselves measured at $t - 1$. Each regression uses the conditioning set of the main model, absorbs agent and week fixed effects, and conditions on the other instruments (Danieli et al., 2026). Standard errors are clustered by agent.

Figures A1 to A4 report the coefficient on each instrument, with its 95 percent confidence interval, for every negative control and lag. One of 20 coefficients is significant for the customer-complicatedness instrument, 1 of 15 for the CRM instrument and 3 of 15 for the GenAI instrument. The product-complicatedness instrument shows more associations, 7 of 20. All coefficients are small, below 0.08 in absolute value, and no association persists across all four lags of any measure. A live alternative path would show as a sustained association on a given measure, and that does not appear. Negative controls are proxies. Small coefficients do not rule out an alternative path, and significant ones may reflect functional-form sensitivity and not a violation (Danieli et al., 2026). Section A3.3 reports a test of the exclusion restriction itself.

A3.3 Control-Function Exclusion Restriction Tests

The negative-control tests in the previous section are indirect: they test whether the instruments are associated with variables they should not affect, not whether the exclusion restriction itself holds. D’Haultfœuille et al. (2024) provide a test of the exclusion restriction itself, available when an instrument satisfies a local irrelevance condition: for some agents, a change in the instrument does not change the endogenous variable.⁹ At that point, any remaining effect of the instrument on performance must run through some channel other than the endogenous variable, which the exclusion restriction rules out.

Testing this requires two steps. First, we confirm local irrelevance holds, using two diagnostics: a conditional-scale test, checking whether the instrument changes the dispersion of the endogenous variable, and a standardized-residual distribution test, checking whether the residual distribution is stable once that scale is accounted for.¹⁰ Second, conditional on local irrelevance, we test the exclusion restriction by comparing the outcome distributions at that point across different

⁹Formally, local irrelevance requires an interior value x^* of the endogenous variable X such that the conditional distribution of X given the instrument Z and controls W , evaluated at x^* , is the same for two distinct instrument values z and z' , with this common value strictly between 0 and 1 (D’Haultfœuille et al., 2024, Assumption 4). The testable implication for the exclusion restriction follows from their Theorem 1.

¹⁰In a location-scale model, this condition holds if and only if the instrument’s effect is heteroskedastic (D’Haultfœuille et al., 2024, Example 1); our conditional-scale and standardized-residual tests operationalize this implication.

instrument values. [Table A3](#) reports both steps. The conditional-scale tests are significant for all four endogenous variables, and none of the standardized-residual tests rejects distributional stability, confirming local irrelevance for *CRM Use*, *GenAI Use*, *Customer Complicatedness*, and *Product Complicatedness*. Conditional on this, the exclusion-restriction test comparing outcome distributions fails to reject its null for all four variables. Because the test is joint over the control-function assumptions and the exclusion restriction, and maintains instrument independence and first-stage monotonicity rather than testing them, this is falsification evidence consistent with exclusion.

A3.4 Shift-Share Instrument Diagnostics

CRM- and GenAI-use identification follows the exogenous-shifts shift-share route, which places the identifying burden on the module-week shocks and not on the exposure shares. This section reports the route's two testable requirements, shock dispersion⁵ and shock independence⁶, as diagnostics on the shock series, independent of the second-stage estimates.

Dispersion. The first requirement is that identification draw on many, sufficiently dispersed shocks and not on a small number of module-weeks carrying most of the identifying weight ([Borusyak et al., 2022](#)). [Table A4](#) reports, following [Borusyak et al. \(2022\)](#), the distribution of each shock series across module-weeks, its concentration via the effective number of shocks (the inverse Herfindahl index of exposure weights), and the largest single module-week weight. The CRM and GenAI shocks have effective sample sizes of 166.79 and 103.72 respectively, out of 1,820 module-week cells, with largest weights of 1.3 and 1.8 percent, which supports the many-shocks condition the shift-share asymptotics require.

Independence. The second requirement is that the shocks be sufficiently uncorrelated, or that any dependence be structured and understood ([Borusyak et al., 2022](#)). Software systems, including CRM platforms, are nearly decomposable, meaning modules interact weakly with the rest of the platform but more tightly within their own functional subsystem ([Simon, 1962](#); [Tiwana & Safadi, 2026](#); [Widjaja & Gregory, 2020](#)). A shock to the platform can therefore act at two layers, a module-group layer shared by every module serving the same function, and a module-specific layer left over once that shared component is removed. Panel A of [Table A5](#) decomposes shock variance along this structure, and in both series the module-specific component carries at least as much of the variance as the module-group component.

The same requirement applies across the two shock series. The CRM and GenAI instruments are built from different sources, module dependency load measured on the platform in the first case and qualifying OpenAI product releases in the second, and the comparison between deterministic and probabilistic feature use holds only where those two series carry separate information. Panel B of [Table A5](#) regresses each series on the other at the module-week level with module and week

fixed effects and standard errors clustered by module, so the quantity tested is the residual variation that reaches agents through the exposure shares. Neither shock predicts the other, and the partial correlation between the two series is -0.012 ($p = .60$). The two instruments therefore identify the two technologies separately.

A3.5 Exposure-Robust Standard Errors

[Table A2](#) reports standard errors clustered on agent. That addresses correlation in an agent's own residuals over time. It leaves correlation across agents, which for a shift-share instrument is the correlation that matters. [Adão et al. \(2019\)](#) show that residuals in shift-share regressions are correlated across units with similar shares, independently of the groupings on which standard errors are conventionally clustered, so conventional estimators are biased downward whenever units with similar shares also have similar residuals. Two agents who draw heavily on the same workflow modules are exposed to the same module-week shocks, so the condition holds here. [Table A6](#) recomputes the first-stage standard errors on the two shift-share instruments using the exposure-robust estimator those authors derive, with and without clustering by module.

The coefficients are unchanged, since the correction bears only on the standard errors. Both instruments remain significant under the exposure-robust estimator and under its module-clustered variant, so the first stage holds under inference that allows for correlation across agents with similar exposure.

The estimator is derived under conditions that require many shocks, each asymptotically small and independent across sectors ([Adão et al., 2019](#)). [Table A5](#) reports the dependence structure of the shock series, and [Section A3.6](#) reports inference that holds without those conditions.

A3.6 Randomization Inference on the GenAI Shock

The GenAI shock is built from R qualifying releases. Release r reaches the module set $\mathcal{M}_r$, where the platform deployed the capability it expands, and becomes active in week t_r , the first full week after its public release date. Collecting these as the assignment $a = \{(\mathcal{M}_r, t_r)\}_{r=1}^R$, the shock and the instrument of [Section 4.3](#) are functions of a :

$$g_{kt}(a) = \sum_{r=1}^R \mathbf{1}\{k \in \mathcal{M}_r\} \mathbf{1}\{t_r \leq t\}, \quad z_{it}(a) = \sum_{k \in \mathcal{K}} w_{ik,t-1}^{\text{type}} g_{kt}(a) \quad (4)$$

Because a was a realised shock, we test whether the first stage reflects the realised assignment or would arise from any assignment with the same structure. Let $\mathcal{K}^*$ be the modules in which the platform deployed a GenAI facility during the window, and $\mathcal{T}^*$ the weeks in which a qualifying release became active. We draw $L = 1,000$ placebo assignments independently and uniformly

from $\mathcal{A} = \{a' = \{(\mathcal{M}'_r, t'_r)\}_{r=1}^R : \mathcal{M}'_r \subseteq \mathcal{K}^*, |\mathcal{M}'_r| = |\mathcal{M}_r|, t'_r \in \mathcal{T}^* \forall r\}$; rebuild $g_{kt}(a^{(l)})$ and $z_{it}(a^{(l)})$ by (A1), and re-estimate the first stage to obtain $\hat{\pi}^{(l)}$. The shares, *GenAI Use*, the controls and the fixed effects are held at their observed values, so each placebo differs from the realised instrument only through g_{kt} . Under the null that *GenAI Use* does not respond to the pairing of releases with modules and weeks, $\hat{\pi}$ and $\hat{\pi}^{(l)}$ share one distribution, and

$p = \frac{1 + \sum_{l=1}^L \mathbf{1}\{|\hat{\pi}^{(l)}| \geq |\hat{\pi}|\}}{L+1}$ is exact in finite samples and bounded away from zero (Phipson & Smyth, 2010).

Figure A5 plots the distribution of $\hat{\pi}^{(l)}$ against the realised coefficient $\hat{\pi} = 0.064$ from Table A2. The placebo coefficients are centred at 0.0006 with sd at 0.032. A shock with the same number of releases, reach and activation weeks, but randomly paired with modules and weeks, produces no first stage. The GenAI first stage in Table A2 therefore does not arise from the exposure shares or from the structure of the release schedule, which is the property the exogenous-shifts design requires (Borusyak & Hull, 2023; Borusyak et al., 2022). The realised coefficient lies at $p = 0.056$ in the placebo distribution. Under exposure-robust inference, the first stage remains significant (Table A6).

A3.7 Mechanical Correlation Between the Count and Portfolio Size

The inverted-U prediction is a statement of concavity, so the quadratic term is the coefficient most exposed to how complicatedness is counted, and an additive control for portfolio size does not remove that exposure because the dependence of the count on size is itself concave.

We test this by permutation. Agent-weeks are grouped into strata by portfolio size, and within each stratum complicatedness and its instrument are reassigned together across the agent-weeks of the estimation sample before column 6 of Tables 4 to 5 is re-estimated. The null is that within a stratum of portfolio size, complicatedness is exchangeable across agent-weeks with respect to conversion, so that an agent's own complicatedness is uninformative about that agent's own conversion beyond what portfolio size implies. Reassignment within strata preserves the relationship between portfolio size and the count and leaves conversion unaltered, so that any association transmitted through portfolio size is retained under the null while the agent-specific association is removed. The placebo distribution is therefore the sampling distribution of the coefficient when the only operative relationship is the dependence of the count on portfolio size, and we draw five hundred replications for each dimension.

Figure A6 reports the result for each of the six coefficients involving complicatedness on each dimension, one panel per coefficient, with the two columns giving *Customer* and *Product Complicatedness*. Each panel plots the distribution of the five hundred placebo coefficients, the observed estimate from column 6, and the placebo mean, which estimates the component of the observed estimate attributable to how complicatedness is counted. The permutation p-value printed

on each panel is the statistic defined in [Section A3.6](#), computed over five hundred replications, so the smallest value it can return is 0.002 and never zero ([Phipson & Smyth, 2010](#)). No placebo replication attained curvature of the observed magnitude on either dimension, and the same holds for the moderation of complicatedness by *CRM Use* on both dimensions, for its quadratic form on *Customer Complicatedness*, and for the moderation by *GenAI Use* on both dimensions. The test addresses only how complicatedness is measured. Within that scope, the inverted U and the associated pattern of moderation are not artefacts of its construction.

A3.8 Robustness to Potentially Post-Treatment Controls

Three of the controls in the structural equation could be consequences of the variables they sit alongside. Process activity burden counts transactions and activities recorded in the CRM, so it may be produced by *CRM Use*, which would make it a mediator and the estimate on *CRM Use* a direct effect. Portfolio age and portfolio value are predicted by the instruments at some lags ([Section A3.2](#)), which places them downstream of the variation the design uses for identification. Conditioning on a consequence of the treatment or of the instrument biases the terms it is meant to leave alone.

We re-estimate Column 6 of the main results twice. The first variant removes all three controls. The second replaces them with their values at $t - 1$, which keeps them in the specification while placing them before the period in which the outcome is measured. Both variants and the baseline run on one common sample, the rows for which the lagged values exist, so that no difference between columns reflects a difference in the observations. The control-function terms are rebuilt inside each column, because the first stages themselves contain these controls and residuals carried from a first stage that still includes a dropped variable would be incoherent. [Table A7](#) reports the result.

No coefficient changes sign, and none changes conventional significance, across both complicatedness dimensions and both variants. The moderation of complicatedness by *CRM Use* is negative and significant throughout, its quadratic form is positive and significant on *Customer Complicatedness* and absent on *Product Complicatedness* in every column, and the moderation by *GenAI Use* is positive and significant throughout. Magnitudes move away from zero when the three controls are removed, by at most a quarter. That is the direction a mediator implies, since a variable absorbing part of the effect should raise the estimate once it is removed, so the published specification is the conservative one. The baseline column is estimated on the common sample and so differs slightly from the published column 6, which uses all agent-weeks. Column 6 reproduces on the full sample before the common sample is imposed.

Table A2. First-Stage Instrument Relevance

	Product Complicatedness (1)	Customer Complicatedness (2)	GenAI Use (3)	CRM Use (4)
GenAI shift-share IV	-0.023** (0.012)	-0.025* (0.014)	0.064*** (0.012)	0.557*** (0.024)
CRM shift-share IV	0.008 (0.008)	0.014 (0.009)	0.005 (0.007)	-0.146*** (0.013)
Product Complicatedness IV	-0.079*** (0.017)	0.045** (0.021)	-0.012 (0.012)	-0.012 (0.016)
Customer Complicatedness IV	0.017 (0.018)	-0.115*** (0.023)	-0.001 (0.011)	-0.026 (0.016)
Constant	-0.132 (0.092)	-0.180 (0.113)	-0.356*** (0.082)	1.751*** (0.158)
Observations	16,408	16,408	16,408	16,408
Controls	Yes	Yes	Yes	Yes
Agent/Time FEs	Yes/Yes	Yes/Yes	Yes/Yes	Yes/Yes
R-squared	0.783	0.715	0.481	0.812
SW F-statistic	50.62	56.98	10.05	11.67

Notes: Standard errors clustered by agent in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A3. Final Local-Irrelevance and Exclusion-Restriction Results

Endogenous variable	Conditional-scale test (1)	Standardized- Residual distribution test (2)	Local-irrelevance condition (3)	Exclusion Restriction test (4)	Exclusion Conclusion (5)
CRM Use	F=35.85***	KS=2.120 (p=.366)	Supported	KS=1.041 (p=.858)	Supported
GenAI Use	F=3.96***	KS=1.287 (p=.792)	Supported	KS=5.676 (p=.846)	Supported
Customer complicatedness	F=69.98***	KS=1.993 (p=.148)	Supported	KS=1.346 (p=.776)	Supported
Product complicatedness	F=241.38***	KS=1.491 (p=.345)	Supported	KS=0.640 (p=.938)	Supported

Table A4. Shift-Share Shock Summary Statistics

	CRM shock (1)	GenAI shock (2)
Mean	11.88	2.81
Standard deviation	2.19	1.97
Interquartile range	3.48	3.00
Effective number of module-week shocks (1/HHI)	166.79	103.72
Largest module-week shock weight	0.013	0.018
Number of module-week shocks	1,820	1,820
Number of CRM modules	65	65
Number of weeks	28	28

Table A5. Shift-Share Shock Dependence Structure

	CRM shock (1)	GenAI shock (2)
Panel A. Within-series variance decomposition (intraclass correlations)		
Module group	0.249 (0.176)	0.373 (0.085)
Module	0.281 (0.086)	0.410 (0.069)
Residual	0.046 (0.044)	0.199 (0.030)
Panel B. Between-series dependence		
GenAI shock	-0.011 (0.040)	-
CRM shock	-	-0.014 (0.052)
Partial correlation between the two series	-0.012 (p=0.60)	
Number of modules	65	65
Number of weeks	30	28
Number of module-weeks	1,950	1,820

Table A6. First-Stage Exposure-Robust Standard Errors

	GenAI Use (1)	CRM Use (2)
Instrument coefficient	0.064***	-0.146***
cluster(agent),	(0.012)	(0.013)
exposure-robust (AKM/BHJ)	(0.018)	(0.013)
exposure-robust, cluster(module)	(0.017)	(0.019)
Observations	16,408	16,408
Controls	Yes	Yes
Agent and Time FEs	Yes/Yes	Yes/Yes
R-squared	0.481	0.812

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors in parentheses. Each row reports a different variance estimator for the same coefficient.

Table A7. Robustness to Potentially Post-Treatment Controls

	Baseline (1)	Parsimonious (2)	Lagged controls (3)
Panel A. Customer complicatedness			
Complicatedness	0.553*** (0.059)	0.563*** (0.060)	0.614*** (0.058)
Complicatedness ²	-0.181*** (0.031)	-0.164*** (0.033)	-0.170*** (0.033)
CRM Use × Complicatedness	-0.137*** (0.031)	-0.160*** (0.033)	-0.162*** (0.033)
CRM Use × Complicatedness ²	0.063*** (0.019)	0.073*** (0.021)	0.072*** (0.021)
GenAI Use × Complicatedness	0.149** (0.068)	0.165** (0.072)	0.174** (0.072)
GenAI Use × Complicatedness ²	0.013 (0.032)	0.022 (0.033)	0.021 (0.033)
Panel B. Product complicatedness			
Complicatedness	0.964*** (0.083)	1.120*** (0.085)	1.194*** (0.086)
Complicatedness ²	-0.051*** (0.018)	-0.052*** (0.019)	-0.055*** (0.019)
CRM Use × Complicatedness	-0.148*** (0.037)	-0.162*** (0.040)	-0.164*** (0.040)
CRM Use × Complicatedness ²	0.008 (0.014)	0.012 (0.015)	0.012 (0.015)
GenAI Use × Complicatedness	0.284*** (0.093)	0.285*** (0.099)	0.297*** (0.099)
GenAI Use × Complicatedness ²	-0.013 (0.014)	-0.007 (0.016)	-0.009 (0.015)
Process activity burden	Yes	No	t-1
Portfolio value	Yes	No	t-1
Portfolio age	Yes	No	t-1
Remaining controls	Yes	Yes	Yes
Agent and Time FEs	Yes/Yes	Yes/Yes	Yes/Yes
Observations	15,822	15,822	15,822

Notes: Bootstrap standard errors clustered on agent. Stars: * p<.10, ** p<.05, *** p<.01. Control-function terms are rebuilt within each column.

Table A8. Full Control-Function Results

	Focal: Customer complicatedness (1)	Focal: Product complicatedness (2)
Endogeneity correction (CF residuals)		
$\hat{v}_{\text{Complicatedness}}$	-0.244*** (0.061)	-0.373*** (0.074)
$\hat{v}_{\text{GenAI Use}}$	1.238 (1.337)	1.204 (1.373)
$\hat{v}_{\text{CRM Use}}$	0.030 (0.139)	0.031 (0.141)
$\hat{v}_{\text{GenAI Use}} \times \text{CRM Use}$	-0.017 (0.060)	-0.007 (0.053)
$\hat{v}_{\text{CRM Use}} \times \text{GenAI Use}$	0.017 (0.037)	0.025 (0.039)
$\hat{v}_{\text{CRM Use}} \times \text{Complicatedness}$	0.070*** (0.025)	0.092*** (0.028)
$\hat{v}_{\text{Complicatedness}} \times \text{CRM Use}$	-0.017 (0.022)	-0.028 (0.022)
$\hat{v}_{\text{GenAI Use}} \times \text{Complicatedness}$	-0.149** (0.065)	-0.260*** (0.088)
$\hat{v}_{\text{Complicatedness}} \times \text{GenAI Use}$	-0.0004 (0.026)	-0.008 (0.029)
N	16408	16408
Adj. R ²	0.684	0.681
Agent FE/Week FE/Controls	Yes/Yes/Yes	Yes/Yes/Yes
Overall F	109.20	108.54
DWH test: Level Terms	18.19***	19.53***
DWH test: Moderation Terms	2.19**	3.33***
DWH test: Joint	8.72***	10.27***

Notes: Bootstrap standard errors in parentheses for coefficient rows. Stars: * p<.10, ** p<.05, *** p<.01. Controls comprise the non-focal complicatedness dimension, product and *Customer Complicatedness* at qualification, process activity burden, portfolio value, portfolio age, bundling, portfolio size, and a constant.

Figure A1. Negative-control outcomes, Customer Complicatedness instrument

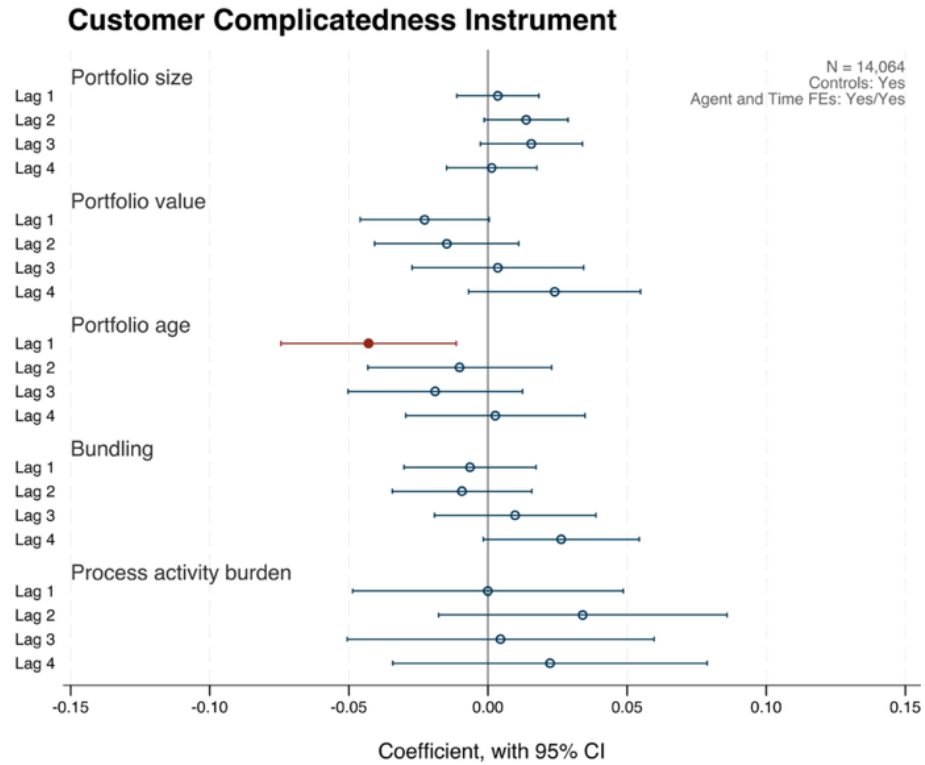


Figure A2. Negative-control outcomes, Product Complicatedness instrument

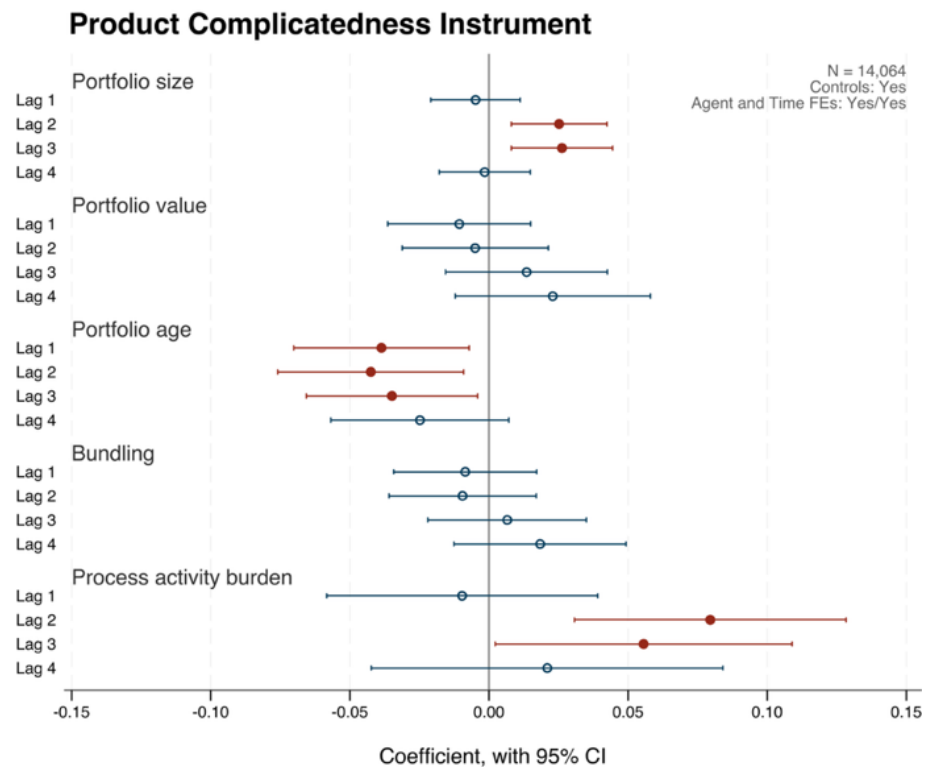


Figure A3. Negative-control outcomes, GenAI shift-share instrument

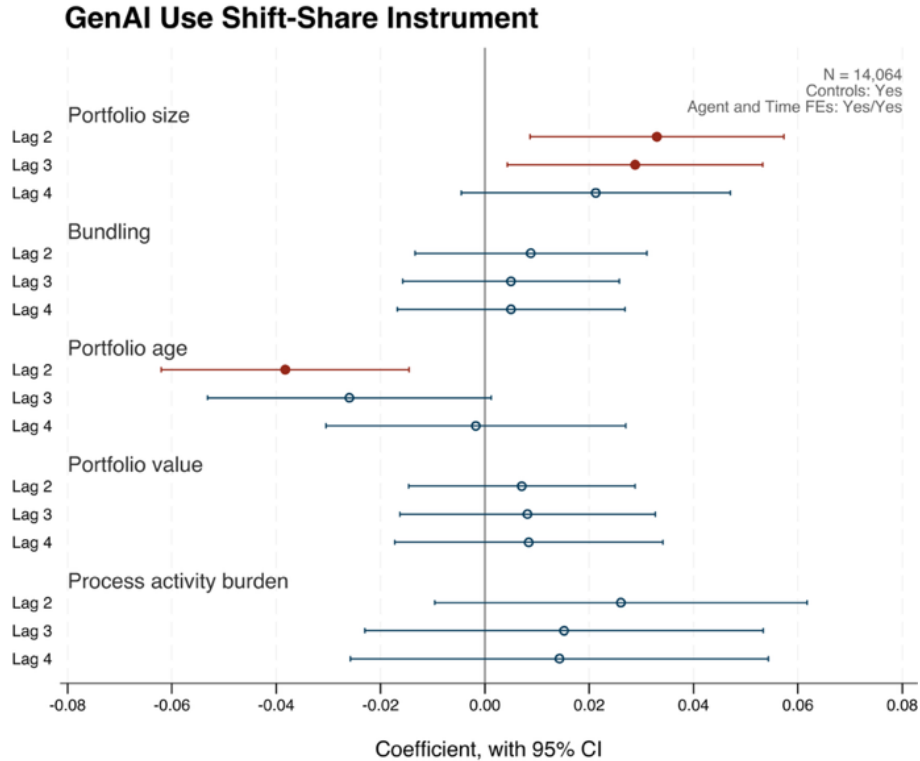


Figure A4. Negative-control outcomes, CRM shift-share instrument

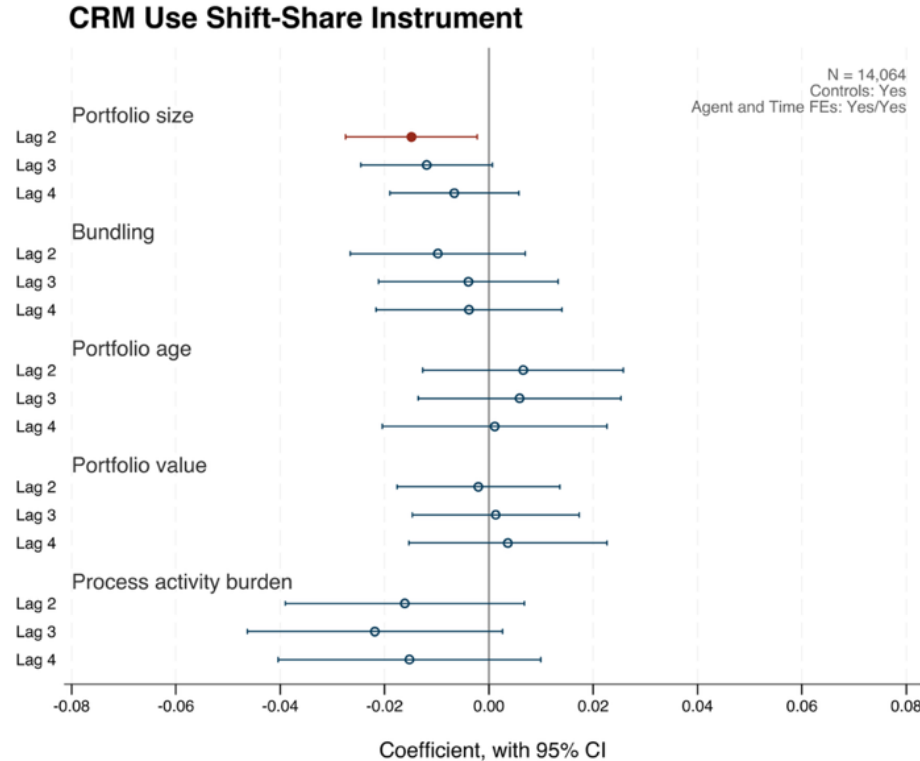
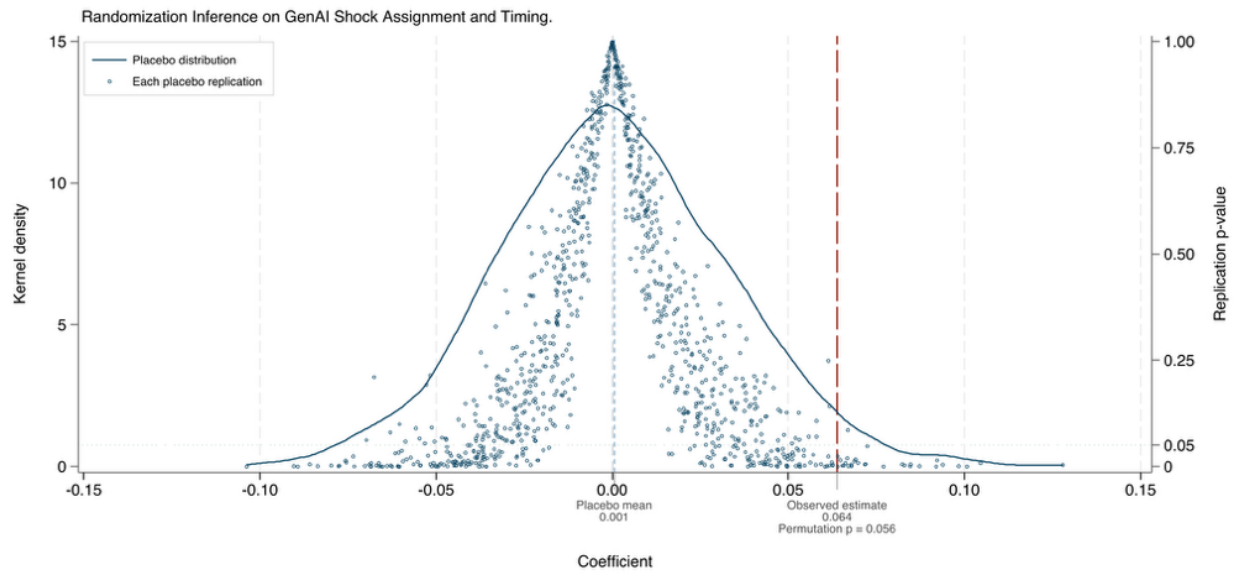
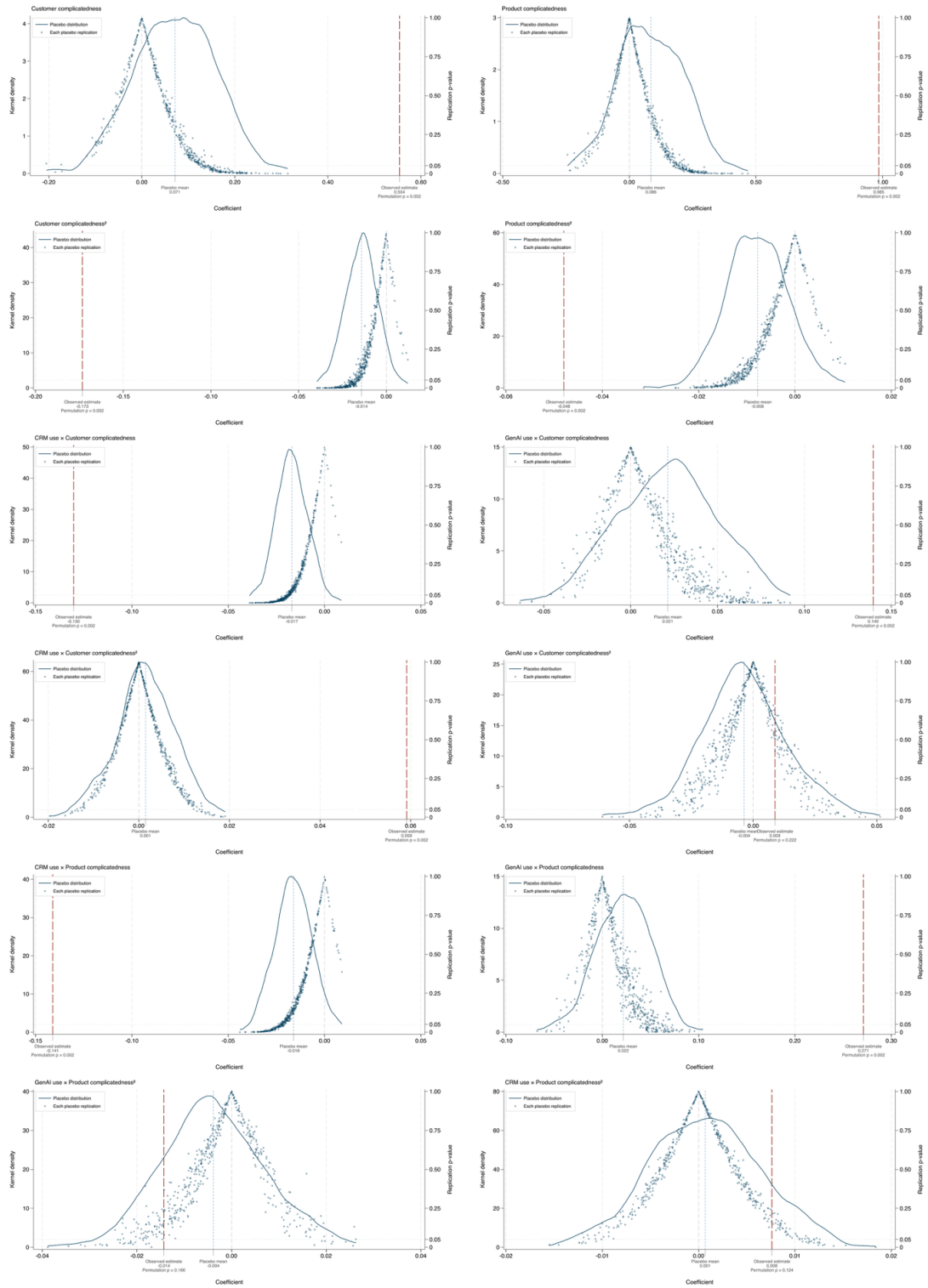


Figure A5. Randomization inference on the assignment and timing of the GenAI shock



Note. Distribution of GenAI first-stage coefficients under 1,000 placebo shocks that redraw which eligible modules each qualifying release reaches and in which activation week it becomes active. The dashed line marks the placebo mean of 0.0007, and the solid line marks the realized coefficient of 0.064 from [Table A2](#), with $p = 0.056$

Figure A6. Permutation test for mechanical correlation, Customer and Product Complicatedness



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